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3 DML-IVQR Methodology

3.1 Structural Quantile Model

Following the instrumental-variable quantile regression (IVQR) framework of Chernozhukov and Hansen (2005) and Chernozhukov and Hansen (2008), let Y_i denote a scalar outcome, D_i an endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. For each potential treatment state d, the potential outcome satisfies

$$Y_{di} = q(U_{di}, d, X_i), \qquad U_{di} \mid X_i \sim \text{Uniform}(0, 1), \tag{1}$$

where U_{di} is the treatment-state-specific structural rank and $q(u, d, x)$ is strictly increasing in u. Thus, $q(\tau, d, x)$ is the conditional τ-quantile of the potential outcome Y_d given $X = x$, and hence the structural quantile obtained by fixing treatment at d. The IVQR assumptions combine conditional instrument exogeneity for the potential-outcome ranks, a treatment-selection mechanism, and rank similarity across treatment states. Define the realized structural rank as $U_i := U_{D_i i}$. Rank similarity does not impose the stronger special case of rank invariance, under which $U_{di} = U_{d'i}$ for all treatment states d and d'. A central implication of these assumptions is the observed-data representation

$$Y_i = q(U_i, D_i, X_i), \qquad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1). \tag{2}$$

Accordingly, U_{di} indexes the rank associated with potential treatment state d, whereas U_i is the rank realized under the observed treatment D_i; the model does not require an individual to retain an identical rank under every treatment state.

Endogeneity permits D_i to be statistically dependent on U_i. Consequently, the observed conditional quantile $Q_{Y|D,X}(\tau \mid d, x)$ generally differs from the structural quantile $q(\tau, d, x)$: the former reflects selection into the realized treatment, whereas the latter holds treatment fixed. Conditioning directly on $D_i = d$ can change the distribution of latent ranks represented among observations and thus need not recover the corresponding potential-outcome quantile. For two treatment states d_1 and d_0, the structural quantile treatment effect at controls x is $q(\tau, d_1, x) - q(\tau, d_0, x)$. The contrast compares the τ-quantiles of the two conditional potential-outcome distributions at common controls. The distinction between observed and structural quantiles is fundamental to IVQR.

For a fixed $\tau \in (0,1)$, the DML-IVQR procedure of Chen et al. (2021) adopts the linear-in-parameters structural quantile specification

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau). \tag{3}$$

Here $c_0(\tau)$ is the quantile-specific intercept, $\alpha_0(\tau)$ is the low-dimensional structural parameter of interest, and $\beta_0(\tau)$ contains the quantile-specific slopes on the nonconstant controls. In the high-dimensional setting, $\beta_0(\tau)$ may have many components and is treated as a nuisance parameter. The dimension of $\alpha_0(\tau)$ equals the number of endogenous variables but is assumed small relative to the potentially large control dimension. If D_i is scalar, $\alpha_0(\tau)$ is the quantile-specific structural treatment effect per unit change in D_i under this linear specification.

Strict monotonicity in the rank gives $\{Y_i \le q(\tau, D_i, X_i)\} = \{U_i \le \tau\}$. Together with the conditional uniformity in (2), this yields the key restriction at the true structural parameters,

$$\Pr\big[Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau) \mid X_i, Z_i\big] = \tau. \tag{4}$$

Equivalently, the model implies the conditional moment

$$\mathrm{E}\big[\tau - \mathbb{1}\{Y_i \le c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau)\} \bigm| X_i, Z_i\big] = 0. \tag{5}$$

Thus, despite the endogeneity of D_i, the rank and instrument-exogeneity structure supplies a conditional quantile restriction for the structural coefficients: at their true values, the probability of a nonpositive structural residual is τ within cells defined by (X_i, Z_i). This restriction defines the population condition on which the later procedure is based. Directly estimating the low-dimensional target $\alpha_0(\tau)$ jointly with the high-dimensional nuisance $\beta_0(\tau)$ is difficult; the next subsection therefore profiles the nuisance coefficient for each candidate treatment parameter.

3.2 Profiled Nuisance Parameters

Fix $\tau \in (0,1)$ and let a denote a candidate value for the low-dimensional structural coefficient $\alpha_0(\tau)$. Define the augmented control vector $\widetilde{X}_i = (1, X_i')'$, whose first element represents the intercept and whose remaining elements are the nonconstant controls. Profiling temporarily treats a as fixed. After subtracting $D_i'a$ from Y_i, the intercept and control slopes are chosen together to fit the τ-quantile of the remaining outcome according to the quantile check loss

$$\rho_\tau(u) = u\big[\tau - \mathbb{1}\{u < 0\}\big]. \tag{6}$$

The loss gives asymmetric linear weights to positive and negative residuals, with the asymmetry determined by τ. Following Chen et al. (2021), the candidate-specific profiled intercept and slopes are

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg\min_{c,b} \mathrm{E}\left[\rho_\tau\left(Y_i - D_i'a - c - X_i'b\right)\right]. \tag{7}$$

Thus, $c(a)$ and $\beta(a)$ are the population intercept and control slopes for the adjusted outcome $Y_i - D_i'a$. Because this outcome changes with a, both quantities can change as well. Under the usual quantile-regression regularity conditions, they have the joint population moment characterization

$$\mathrm{E}\left[\left\{\tau - \mathbb{1}\left[Y_i \le D_i'a + c(a) + X_i'\beta(a)\right]\right\}\widetilde{X}_i\right] = 0. \tag{8}$$

The constant element of $\widetilde{X}_i$ gives the intercept condition, while the remaining elements give the slope conditions for the nonconstant controls. Together, these moments define the intercept and control coefficients associated with candidate a. At $a = \alpha_0(\tau)$, the maintained model and uniqueness conditions imply $c(\alpha_0(\tau)) = c_0(\tau)$ and $\beta(\alpha_0(\tau)) = \beta_0(\tau)$. For $a \neq \alpha_0(\tau)$, however, $c(a)$ and $\beta(a)$ are candidate-specific profile nuisances rather than the true structural intercept and control coefficients.

The associated profile residual is

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \tag{9}$$

Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ denote its conditional density evaluated at zero. Residuals below zero lie on one side of the fitted candidate quantile, while residuals above zero lie on the other. A small change in $c(a)$ or $\beta(a)$ can move observations with residuals close to zero from one side to the other. The density near zero therefore determines how sensitive the quantile moment is to small changes in the fitted intercept and control slopes.

To characterize this sensitivity, suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\widetilde{X}_i \in \mathbb{R}^{p+1}$, and define

$$\begin{aligned} M(a) &= \mathrm{E}\left[Z_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\right], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathrm{E}\left[\widetilde{X}_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\right], \quad J(a) \in \mathbb{R}^{(p+1)\times(p+1)}. \end{aligned} \tag{10}$$

The matrices summarize how the quantile moment changes when the fitted intercept or control slopes move slightly. The matrix $J(a)$ describes this relationship using the augmented controls themselves, whereas $M(a)$ describes the corresponding relationship between the instruments and the augmented controls. Both are weighted by the residual density at zero because observations near the fitted quantile are most likely to change sides after such a movement. Assuming that $J(a)$ is nonsingular at the population level, set

$$\delta(a) = M(a)J(a)^{-1}, \qquad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \tag{11}$$

Partition this matrix as $\delta(a) = [d_0(a), \Delta(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\Delta(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\widetilde{X}_i = Z_i - d_0(a) - \Delta(a)X_i. \tag{12}$$

The matrix $\delta(a)$ determines how much of each instrument is associated with the intercept and controls under the density weighting. Hence, $\Psi_i(a)$ is the part of the instrument left after removing that weighted component. By construction, $\mathrm{E}[(Z_i - \delta(a)\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$. Because the weighting gives more importance to observations near the candidate quantile boundary, $\Psi_i(a)$ is not an ordinary unweighted regression residual, and $\delta(a)$ is not a conventional first-stage coefficient.

Collect the candidate-specific nuisance quantities as

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \mathrm{vec}\{\delta(a)\} \end{pmatrix}, \qquad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \mathrm{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \tag{13}$$

Here $\mathrm{vec}\{\delta(a)\}$ stacks the entries of the matrix $\delta(a)$ into one vector. Vectorization changes only how the matrix nuisance is stored; it does not alter the residualization itself. These objects are nuisance quantities because the inferential target remains $\alpha_0(\tau)$. Candidate-specific $c(a)$ and $\beta(a)$ handle the structural intercept and high-dimensional control component, while $\delta(a)$ removes the density-weighted component of the instruments associated with $\widetilde{X}_i$. Section 3.3 combines these objects with the IVQR quantile residual to construct a Neyman-orthogonal moment for $\alpha_0(\tau)$.

3.3 Neyman-Orthogonal Score

Following Chen et al. (2021), combine the nuisance quantities from Section 3.2 in the score

$$\begin{aligned} g_\tau(W_i; a, \eta) &= \left[\tau - \mathbb{1}\left\{Y_i \le D_i'a + c + X_i'\beta\right\}\right]\left(Z_i - \delta\widetilde{X}_i\right), \\ g_\tau(W_i; a, \eta(a)) &= \left[\tau - \mathbb{1}\left\{Y_i \le D_i'a + c(a) + X_i'\beta(a)\right\}\right]\Psi_i(a), \end{aligned} \tag{14}$$

where $W_i = (Y_i, D_i, X_i, Z_i)$ collects the observed variables and η contains c, β, and δ. The first factor is the quantile residual, or quantile score. It compares the observed outcome with the fitted structural τ-quantile. It equals τ when the outcome is above the fitted quantile and $\tau - 1$ when the outcome is at or below it; it is therefore not an ordinary regression residual. At the true model, its conditional expectation is zero. The second factor is the residualized instrument from Section 3.2. It removes the part of Z_i associated with the intercept and nonconstant controls under the density-weighted projection and leaves the instrument variation used in the moment. Multiplying these two parts gives the IVQR moment used to assess the candidate value a.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \tag{15}$$

Section 3.1 showed that the true quantile residual has conditional mean zero given X_i and Z_i. The residualized instrument is itself a function of X_i and Z_i. Thus, within every value of these variables, multiplying by the residualized instrument still gives conditional mean zero. Averaging over X_i and Z_i gives (15). This moment is valid because of the structural-rank and instrument-exogeneity assumptions in Section 3.1. Neyman orthogonality is a separate property created by the nuisance construction. It requires

$$\partial_\eta \, \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} = 0. \tag{16}$$

This condition does not mean that the nuisance parameters are known. It says that small errors around their true values do not change the population moment to first order. Without orthogonality, a small nuisance error can cause a first-order change in the target moment. With it, that first-order change is zero, so the leading effect of a sufficiently small error is of higher order. Orthogonality therefore reduces the effect of nuisance-estimation error; it does not remove that error.

To see the joint cancellation for the intercept and control slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$. Using the residual density defined in Section 3.2, define

$$\begin{aligned} M_0 &= \mathrm{E}\left[Z_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\right], \\ J_0 &= \mathrm{E}\left[\widetilde{X}_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\right], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \tag{17}$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned}\partial_{(c,\beta')} \mathrm{E}[g_\tau(W_i;\alpha_0(\tau),\eta)]\big|_{\eta=\eta_0} &= -\,\mathrm{E}\big[(Z_i - \delta_0\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big] \\ &= -\,[M_0 - \delta_0 J_0] = 0.\end{aligned} \tag{18}$$

A small error in c or β moves the fitted quantile. Observations close to the quantile boundary are the ones most likely to switch sides, which is why the density at zero appears. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\widetilde{X}_i$. Its constant entry handles small errors in the fitted intercept, while its remaining entries handle small errors in the control slopes. The corresponding first-order changes therefore cancel.

Orthogonality also holds in the δ direction. For instrument component j, let δ_j denote the corresponding row of δ. Then

$$\partial_{\delta_j} \mathrm{E}\big[g_{\tau,j}(W_i;\alpha_0(\tau),\eta)\big]\big|_{\eta=\eta_0} = -\,\mathrm{E}\big[\big\{\tau - \mathbb{1}\big[Y_i \le D_i'\alpha_0(\tau) + c_0(\tau) + X_i'\beta_0(\tau)\big]\big\}\,\widetilde{X}_i'\big] = 0. \tag{19}$$

A small error in δ changes how much of $\widetilde{X}_i$ is removed from an instrument. That error is multiplied by the quantile residual, whose population moment with $\widetilde{X}_i$ is already zero at $c_0(\tau)$ and $\beta_0(\tau)$ by (8). Thus, it has no first-order effect either. The constant entry covers the residualization-intercept direction, while the remaining entries cover the residualization slopes. This protection matters because high-dimensional methods estimate the nuisance quantities imperfectly, and regularization adds estimation and shrinkage error. The orthogonal score allows sufficiently accurate nuisance estimates to be combined later with inference on the low-dimensional target; this is the specific DML-IVQR construction of Chen et al. (2021) and the general principle explained by Chernozhukov et al. (2018).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a. IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.4 Nuisance Estimation and Regularization

Sections 3.2–3.3 defined $c(a)$, $\beta(a)$, and $\delta(a)$ as population quantities. Their sample counterparts can be difficult to estimate when the number of controls is large relative to the sample size N. Estimating every coefficient without restrictions may then be noisy or unstable. The DML-IVQR procedure therefore uses ℓ_1 regularization. This approach is suited to approximate sparsity: many controls may be available, but only a relatively small group has a large predictive role, while many other coefficients are zero or small. Approximate sparsity does not require every weak or irrelevant coefficient to be exactly zero.

Fix τ and a candidate value a. Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D_i'a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \dots, b_p)'$ contain only the slopes on X_i. Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$, while the later pair $(\widehat{d}_0(a), \widehat{\delta}(a))$ corresponds to the intercept-and-slope partition $[d_0(a), \varDelta(a)]$ of the population residualization matrix. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \arg\min_{b_0, b} \left\{ \sum_{i=1}^{N} \rho_\tau\big(Y_i - D_i'a - b_0 - X_i'b\big) + \lambda_0(a)|b_0| + \sum_{k=1}^{p} \lambda_k(a)|b_k| \right\}. \tag{20}$$

The first term measures fit at quantile τ. The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum because this is the scaling used by the fitted quantile-regression routine. Following the pivotal route described by Chen et al. (2021), let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \dots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The implemented penalty is

$$\begin{aligned} T_r &= \max_{1 \le k \le p} \left| \frac{\sum_{i=1}^{N} X_{ik} \left[\tau - \mathbb{1}\{U_{ir} < \tau\}\right]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda(a) &= 2\, \widehat{Q}_{0.9}\Big(\{T_r\}_{r=1}^{1000}\Big) \sqrt{\tau(1-\tau)}(1, s_1, \dots, s_p)'. \end{aligned} \tag{21}$$

Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned}\widehat{\epsilon}_i(a) &= Y_i - D_i'a - \widehat{b}_0(a) - X_i'\widehat{\beta}(a),\\ \widehat{f}_i(a) &= \phi\left(\widehat{\epsilon}_i(a); \overline{\widehat{\epsilon}}(a), s^2_{\widehat{\epsilon}(a)}\right),\end{aligned} \tag{22}$$

where $\phi(e;\mu,\sigma)$ is the normal density with mean μ and standard deviation σ. Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate residuals.[1] This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component j, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{d}_{j0}(a)$ denote its intercept and let $\widehat{\delta}_j(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix}\widehat{d}_{j0}(a)\\ \widehat{\delta}_j(a)\end{pmatrix} = \underset{(d_{j0},d_j)}{\arg\min}\left\{\sum_{i=1}^{N}\left[\sqrt{\widehat{f}_i(a)}Z_{ji} - d_{j0} - \sqrt{\widehat{f}_i(a)}X_i'd_j\right]^2 + \mathcal{P}_{j,a}(d_j)\right\}. \tag{23}$$

Here $\mathcal{P}_{j,a}(d_j) = \sum_{k=1}^{p}\lambda^{\delta}_{jk}(a)|d_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the intercept and the slopes. Define $\widehat{d}_0(a) = (\widehat{d}_{10}(a), \ldots, \widehat{d}_{d_z0}(a))'$ and stack the slope rows in $\widehat{\delta}(a)$. The residualized instrument used in the score is

$$\begin{aligned}\widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{d}_{j0}(a) - X_i'\widehat{\delta}_j(a),\\ \widehat{\Psi}_i(a) &= Z_i - \widehat{d}_0(a) - \widehat{\delta}(a)X_i.\end{aligned} \tag{24}$$

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The ℓ_1 penalties place a cost on large collections of nonzero coefficients. Weak coefficients can consequently be shrunk strongly, often to zero, which can stabilize both nuisance

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

steps in a high-dimensional sample. This regularization also creates shrinkage error; the orthogonal construction in Section 3.3 is designed to limit its first-order effect on the target moment, as in the general framework of Chernozhukov et al. (2018). The implementation estimates the nuisance parameters and evaluates the score on the same sample; it does not use sample splitting or cross-fitting. General DML procedures often use cross-fitting, but this specific implementation instead relies on the orthogonal score together with its high-dimensional sparsity and rate conditions.

3.5 Oracle-GMM, Full-GMM, and DML-IVQR

The analysis compares three estimators from the IVQR setting studied by Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR all target the same low-dimensional parameter $\alpha_0(\tau)$ in the structural quantile model. For any candidate a, they use the same quantile index, outcome, treatment, and instrument vector Z_i. Each estimator forms a structural quantile residual, residualizes the instruments with respect to a chosen control set, and multiplies the resulting instrument vector by the quantile score. Their candidate-specific moments therefore have the same dimension as Z_i. What differs is how the control coefficient and instrument residualization are estimated.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \ldots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\mathrm{orc}}(a) \\ \widehat{\beta}_{\mathrm{orc}}(a) \end{pmatrix} = \underset{b_0, b}{\arg\min} \sum_{i=1}^{N} \rho_\tau\left(Y_i - D_i' a - b_0 - X_{S_0,i}' b\right). \tag{25}$$

Thus, Oracle-GMM first removes $D_i' a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\mathrm{full}}(a) \\ \widehat{\beta}_{\mathrm{full}}(a) \end{pmatrix} = \underset{b_0, b}{\arg\min} \sum_{i=1}^{N} \rho_\tau\left(Y_i - D_i' a - b_0 - X_i' b\right). \tag{26}$$

In the executable design, this means that all 100 controls enter freely. Full-GMM is feasible in the sense that it does not use the unknown active set, but estimating many weak or irrelevant coefficients from one finite sample can be difficult. Including many controls does not by itself imply inconsistency.

Oracle-GMM and Full-GMM use the same unregularized instrument-residualization calculation, with their respective control matrices. For $e \in \{\text{orc}, \text{full}\}$, let $X_{\text{orc},i} = X_{S_0,i}$ and $X_{\text{full},i} = X_i$. Let $\widehat{f}_{e,i}(a)$ be the fitted-normal density proxy from the estimator's candidate residual, formed with the literal convention described in Section 3.4. The code computes

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} Z_i X'_{e,i}\widehat{f}_{e,i}(a), \qquad & \widehat{J}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} X_{e,i} X'_{e,i}\widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a)\widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}. \end{aligned} \tag{27}$$

This is a direct matrix inversion, not a penalized regression. The residualization includes no intercept, even though the preceding quantile regression does. This is an implementation-specific feature of the frozen benchmark estimators; it differs from the augmented population residualization used to describe the orthogonal DML score in Sections 3.2–3.3. Accordingly, the benchmark moments do not implement the full augmented orthogonality calculation. The calculation also uses the original Z_i in the final residual. The implementation has no ridge adjustment or generalized inverse: if $\widehat{J}_e(a)$ is singular, the direct solve fails and the outer evaluation records a numerical failure. Each benchmark then multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR instead uses the full high-dimensional control set and regularizes both nuisance-estimation steps, as described in Section 3.4. It estimates the control coefficients by quantile Lasso, forms the density proxy, and estimates the instrument residualization through regularized weighted regressions. These nuisance estimates are then used in the score described in Section 3.3. The implemented version, DML-IVQR-BC, uses the pivotal penalty in (21). Oracle-GMM and Full-GMM also use residualized IVQR moments, but their nuisance parameters are estimated without ℓ_1 regularization. Unlike the benchmark matrix calculation, the DML instrument regression retains an intercept. The comparison therefore covers three cases: the oracle control set is known and estimated without regularization; the full control set is estimated without regularization; or the full control set is handled with regularized nuisance estimation. It does not perfectly separate every source of finite-sample error, but it keeps the structural model, target, and instruments common. Once the estimator-specific moment is constructed, Section 3.6 applies the common criterion and inference procedure.

3.6 GMM Criterion, Point Estimation, and Robust Inference

Let $e \in \{\text{orc}, \text{full}, \text{dml}\}$ index the estimator and let $\widehat{g}_{e,i}(a)$ denote its observation-level IVQR score at candidate a. This score uses the estimator-specific nuisance quantities described above and is a $d_z = \dim(Z_i)$ dimensional vector. Following Chen et al. (2021), its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N}\sum_{i=1}^{N}\widehat{g}_{e,i}(a). \tag{28}$$

For a given candidate, this vector shows how far the sample IVQR conditions are from zero. A compatible candidate should produce relatively small average moments after their sampling variation is taken into account. The frozen code measures that variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N}\sum_{i=1}^{N}\widehat{g}_{e,i}(a)\widehat{g}_{e,i}(a)'. \tag{29}$$

The components can have different scales and can move together, so this matrix standardizes them before they are combined. It uses denominator N, has no finite-sample correction, and does not subtract the sample mean. At the true parameter, where the population moment is zero, this uncentered second moment and the covariance have the same probability limit.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N\widehat{g}_{e,N}(a)'\widehat{\Sigma}_e(a)^{-1}\widehat{g}_{e,N}(a). \tag{30}$$

This expression is exactly equivalent to the code, which stores the unnormalized score sum and the unnormalized sum of score outer products. When the second-moment matrix is well behaved, $W_{e,N}(a)$ is nonnegative. A smaller value means that the covariance-weighted moments are closer to zero, while a larger value indicates stronger disagreement between the candidate and the sample restrictions. The criterion combines all instrument components into one number. The code uses a direct matrix inverse without a ridge adjustment or generalized inverse. If inversion fails or the result is non-finite, that candidate is recorded as a numerical failure. When the estimator is clear, the subscript e is suppressed and the statistic is written as $W_N(a)$. Conceptually, the point estimator is

$$\widehat{\alpha}_e(\tau) = \arg\min_{a\in\mathcal{A}} W_{e,N}(a). \tag{31}$$

It selects the candidate that makes the weighted sample moments as close to zero as possible; the minimum need not equal zero. In the Monte Carlo implementation, minimization is numerical over the same prespecified finite profile used for confidence-set evaluation, whose details appear in Section 4.5. The first candidate attaining the minimum is selected if there is an exact tie. An incomplete profile receives no point estimate.

Point estimation and robust inference use the profile differently. Point estimation keeps only its minimizer. Inference instead tests every hypothesized value a separately, so a value need not minimize $W_{e,N}(a)$ to be retained. Under $H_0 : a = \alpha_0(\tau)$, the implemented procedure uses the asymptotic reference result

$$W_{e,N}(a) \xrightarrow{d} \chi^2_{d_z}, \qquad d_z = \dim(Z_i). \tag{32}$$

The degrees of freedom equal the number of instrument components and therefore the number of moment components. Following Chernozhukov and Hansen (2008), this test evaluates the moments directly at the hypothesized value instead of relying on a conventional normal approximation for $\widehat{\alpha}_e(\tau)$. For significance level $q \in (0, 1)$, let c_{1-q,d_z} be the $(1 - q)$ quantile of $\chi^2_{d_z}$. The candidate is rejected when $W_{e,N}(a) > c_{1-q,d_z}$ and is not rejected when the weak inequality holds.

Inverting these candidate-specific tests gives

$$C_{e,1-q}(\tau) = \left\{a \in \mathcal{A} : W_{e,N}(a) \leq c_{1-q,d_z}\right\}. \tag{33}$$

Rather than placing one standard-error band around the point estimate, test inversion retains every candidate that is not rejected by its own moment test. Their collection is the confidence region. Because the profile can cross the critical value more than twice, this region need not be one ordinary interval: it may be connected, have separated components, be wide, or be empty. Numerical evaluation over a finite domain describes only the accepted set within that domain. Acceptance reaching a numerical boundary does not establish that the theoretical region is globally unbounded.

The test-inversion procedure is designed to retain asymptotic validity without requiring strong identification and can accommodate weak, partial, or failed identification under the maintained assumptions. This robustness does not imply that the confidence region will be narrow. With informative instruments, moving a away from the true value tends to move the moments away from zero, so the profile usually rises more clearly. With weak instruments, the moments may change slowly and many candidates may remain below the cutoff. This is the difference between validity and informativeness. Weak-identification robustness concerns asymptotic validity: because the implementation uses an asymptotic chi-square critical value, finite-sample size accuracy remains an empirical question. The method is distinct from the exact finite-sample simulation approach of Chernozhukov et al. (2009).

4 Monte Carlo Design

4.1 Baseline Data-Generating Process

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i, the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \tag{34}$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (34), and they satisfy

$$x_{ji} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \quad j = 1, \ldots, 100, \qquad z_{1i}, z_{2i}, v_{1i}, v_{2i} \overset{\text{ind}}{\sim} \mathcal{N}(0,1). \tag{35}$$

Observations are generated independently across i. The observed controls are transformations of the latent Gaussian draws:

$$X_{ji} = \Phi(x_{ji}), \qquad j = 1, \ldots, 100, \tag{36}$$

where Φ denotes the standard-normal cumulative distribution function. By the probability integral transform, $X_{ji} \sim \mathcal{U}(0,1)$. These transformations preserve independence across controls and map the latent Gaussian draws monotonically onto the unit interval.

The executable design constructs two instruments as[2]

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \tag{37}$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \tag{38}$$

Here, z_{1i} and z_{2i} are the components of the observed instruments that enter the treatment index directly, while the included X components induce dependence between the instruments

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j. The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

and observed controls and v_{1i} and v_{2i} add independent noise. The original baseline treatment equation is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \tag{39}$$

The treatment is continuous and lies in $(0, 1)$. The next subsection introduces the instrument-strength extension by modifying this treatment equation while leaving the remaining structural DGP unchanged.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5 \sum_{j=1}^{7} X_{ji} + D_i u_i. \tag{40}$$

Only $X_{1i}, \ldots, X_{7i}$ enter the structural outcome directly, while $X_{8i}, \ldots, X_{10i}$ additionally enter the instrument construction. The union of the controls appearing in the outcome and instrument equations therefore gives $X_{1i}, \ldots, X_{10i}$ as the ten controls relevant to the complete DGP. For a treatment level $d \in (0, 1)$, Equation (40) implies the potential outcome

$$Y_i(d) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d(1 + u_i). \tag{41}$$

Conditional on X_i, treatment-effect heterogeneity is governed by $1 + u_i$. Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5 \sum_{j=1}^{7} X_{ji} + d \left[1 + \Phi^{-1}(\tau)\right]. \tag{42}$$

The coefficient on d in the structural τ-quantile is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \tag{43}$$

In particular, the median structural quantile treatment effect equals $\alpha_0(0.50) = 1$.

4.2 Instrument-Strength Extension

The thesis-specific extension varies excluded-instrument relevance through a single parameter $\kappa \in [0, 1]$ while holding the remaining structural DGP fixed. It adds one primitive, $w_i \sim \mathcal{N}(0, 1)$, which is independent of all primitives defined in Section 4.1. The latent treatment index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)}\, w_i + \epsilon_i, \tag{44}$$

$$D_i(\kappa) = \Phi\big(d_i^*(\kappa)\big)\,. \tag{45}$$

The common loading κ enters symmetrically on z_{1i} and z_{2i}, so excluded-instrument relevance remains a one-dimensional experimental factor. This treatment equation is the sole structural modification relative to the executable baseline.

At $\kappa = 1$, the coefficient on w_i is $\sqrt{2(1-\kappa^2)} = 0$, and hence

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \tag{46}$$

which exactly reproduces Equation (39). The scaling of w_i preserves the variance of the latent treatment index. Independence and the unit variances of z_{1i}, z_{2i}, w_i, and ϵ_i imply

$$\mathrm{Var}\big[d_i^*(\kappa)\big] = 2\kappa^2 + 2(1-\kappa^2) + 1 = 3. \tag{47}$$

The excluded-instrument component contributes $2\kappa^2$, the added w_i component contributes $2(1-\kappa^2)$, and ϵ_i contributes one. Thus, $d_i^*(\kappa) \sim \mathcal{N}(0,3)$ for every κ, making the marginal distribution of $D_i(\kappa)$ invariant across designs. This distribution is not uniform because Φ is applied to a normal index with variance three rather than one. The added term therefore prevents a mechanical change in overall treatment-index dispersion when the excluded-instrument loading is reduced.

Instrument relevance changes because

$$\mathrm{Cov}\big(Z_{1i}, d_i^*(\kappa)\big) = \mathrm{Cov}\big(Z_{2i}, d_i^*(\kappa)\big) = \kappa. \tag{48}$$

By Equations (37) and (38), the components of each observed instrument other than z_{1i} or z_{2i} are independent of $d_i^*(\kappa)$, whereas $\mathrm{Cov}(z_{1i}, d_i^*(\kappa)) = \mathrm{Cov}(z_{2i}, d_i^*(\kappa)) = \kappa$. Thus, as κ declines, so does the covariance between each observed instrument and the latent treatment index. The parameter κ is the structural loading governing instrument relevance, not a complete measure of IVQR identification. Calibration diagnostics later assess the empirical strength separation of the selected designs. At the same time, endogeneity is preserved because u_i is independent of z_{1i}, z_{2i}, and w_i, whereas its covariance with ϵ_i remains fixed:

$$\mathrm{Cov}\big(u_i, d_i^*(\kappa)\big) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3. \tag{49}$$

Because $(u_i, d_i^*(\kappa))$ is jointly Gaussian with zero means and invariant moments $\mathrm{Var}(u_i) = 1$, $\mathrm{Var}(d_i^*(\kappa)) = 3$, and $\mathrm{Cov}(u_i, d_i^*(\kappa)) = 0.3$, its joint distribution is unchanged across κ. The extension also leaves unchanged the observed instruments in Equations (37) and (38), the controls and their distribution, the disturbance distribution, the outcome in Equation (40), and the structural quantile effect in Equation (43). It is therefore designed to isolate weaker excluded-instrument relevance from changes in the marginal treatment distribution, the endogeneity channel, and the structural treatment-effect distribution.

4.3 Simulation Design

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} &n \in \{500, 1000\}, \qquad &&\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ &\kappa \in \{1, 0.50, 0.25, 0.10\}, &&R = 500. \end{aligned} \tag{50}$$

The two sample sizes characterize finite-sample behavior at different values of n, while the five quantile indices cover the center and both tails of the structural outcome distribution. The instrument-loading values span the executable baseline at $\kappa = 1$ and progressively lower degrees of excluded-instrument relevance; their empirical calibration is presented later. The Cartesian product contains $2 \times 5 \times 4 = 40$ distinct (n, τ, κ) designs, with $R = 500$ Monte Carlo replications in each design cell.

Each design is evaluated using Oracle-GMM, Full-GMM, and DML-IVQR. Oracle-GMM uses the known low-dimensional set of controls relevant to the DGP and serves as an infeasible benchmark. Full-GMM instead uses the complete high-dimensional control set without the DML regularization strategy. DML-IVQR applies the authors' high-dimensional regularized procedure with an orthogonal score. Chapter 3 provides the formal estimator definitions. Combining the 40 designs with the three estimators yields $40 \times 3 = 120$ design-estimator cells, each containing 500 replications. For every quantile index, the true parameter used in the Monte Carlo evaluation is the value from Equation (43), $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$.

Within a given replication and (n, τ, κ) design, the runner evaluates all three estimators on the same generated sample, thereby holding the simulated data fixed across estimator comparisons. The implemented DML-IVQR estimator uses the BC pivotal-penalty and same-sample nuisance-estimation procedure described in Section 3.4.

4.4 Performance Measures

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r, and let $\alpha_0(\tau)$ be the true value from Equation (43). Replications are indexed by $r = 1, \ldots, R$.

Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned}
\mathrm{Bias}(\tau) &= \frac{1}{R}\sum_{r=1}^{R}\left[\alpha_0(\tau) - \widehat{\alpha}_r(\tau)\right], \\
\mathrm{MAE}(\tau) &= \frac{1}{R}\sum_{r=1}^{R}\left|\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right|, \\
\mathrm{RMSE}(\tau) &= \left\{\frac{1}{R}\sum_{r=1}^{R}\left[\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right]^2\right\}^{1/2}.
\end{aligned} \tag{51}$$

Every measure is computed separately for each estimator and design cell; aggregation never pools distinct design cells. Under the frozen reporting convention, positive Bias means that estimates lie below the true parameter on average, while negative Bias means that they lie above it. Bias captures signed systematic deviation, MAE measures absolute error, and RMSE places greater weight on large errors.

Coverage is determined by evaluating the test statistic directly at the exact true parameter. With $c_{0.95}$ denoting the critical value defined in Section 4.5, define

$$C_r(\tau) = \mathbf{1}\left\{W_{N,r}(\alpha_0(\tau)) \le c_{0.95}\right\}, \qquad \widehat{\mathrm{Coverage}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} C_r(\tau). \tag{52}$$

Nominal coverage is 0.95. Crucially, coverage evaluates $W_{N,r}$ at $\alpha_0(\tau)$ itself; it is not inferred from whether a numerical confidence-region grid contains the true value and is therefore independent of whether the truth is a grid point. Empirical size is the rejection frequency at the true parameter:

$$\widehat{\mathrm{Size}}(\tau) = \frac{1}{R}\sum_{r=1}^{R}\mathbf{1}\left\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\right\} = 1 - \widehat{\mathrm{Coverage}}(\tau). \tag{53}$$

The final equality follows exactly from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_\Delta(\tau) = \alpha_0(\tau) + \Delta, \qquad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \tag{54}$$

For each offset, empirical power is

$$\widehat{\mathrm{Power}}(\Delta, \tau) = \frac{1}{R}\sum_{r=1}^{R}\mathbf{1}\left\{W_{N,r}(a_\Delta(\tau)) > c_{0.95}\right\}. \tag{55}$$

The offset $\Delta = 0$ is excluded because rejection at the true value measures size rather than power. As with coverage, the statistic is evaluated directly at $a_\Delta(\tau)$, so power does not depend on whether that false value lies on the numerical confidence-region grid.

Inferential informativeness is summarized by the grid-based accepted-set measure within the prespecified primary computational domain. In each replication, candidate values satisfying $W_{N,r}(a) \le c_{0.95}$ are accepted by the inverted test. The measure totals all accepted components inside the finite domain, including disconnected components; it is neither the distance between the minimum and maximum accepted values nor a claim about the unrestricted theoretical confidence region. Section 4.5 defines the exact numerical grid and trapezoidal construction. The Monte Carlo summary reports the median accepted-set measure across replications. It also reports the all-grid-points acceptance rate, the fraction of replications in which every candidate point in the prespecified primary grid is accepted; this is not acceptance of the entire parameter space. Coverage and size assess empirical calibration at the true parameter, whereas the accepted-set measure and power assess informativeness and discrimination against other parameter values.

4.5 Confidence-Region Construction and Numerical Grid

The Monte Carlo implementation uses the candidate-specific statistic $W_N(a)$ and the uncentered score second-moment matrix defined in Section 3.6. Because the DGP has $d_z = 2$ instrument components, the 95% cutoff is

$$c_{0.95} = \chi^2_{2,0.95} = 5.991464547\ldots. \tag{56}$$

As in Section 3.6, a candidate is accepted when $W_N(a) \le c_{0.95}$. This asymptotic cutoff is not an exact finite-sample guarantee; finite-sample calibration is evaluated empirically in Chapter 5. The main simulation evaluates the accepted set on the prespecified primary grid

$$\begin{aligned} A_0 &= [-1, 3], && h = 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \ldots, 41\}, \\ &= \{-1, -0.9, \ldots, 2.9, 3\}. \end{aligned} \tag{57}$$

The same grid is used across sample sizes, quantiles, instrument-loading values, estimators, and replications. The interval A_0 is solely a computational domain, not the asserted theoretical parameter space. Reaching a boundary of A_0 therefore establishes neither global boundedness nor global unboundedness of the conceptual confidence set.

For replication r, define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\left\{W_{N,r}(a_j) \le c_{0.95}\right\}, \\ C_{r,G_0} &= \left\{a_j \in G_0 : I_r(a_j) = 1\right\}, \\ F_r &= \mathbf{1}\left\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\right\}. \end{aligned} \tag{58}$$

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} \left[I_r(a_j) + I_r(a_{j+1})\right]. \tag{59}$$

This quantity approximates the total measure of accepted values inside A_0. Separated accepted regions contribute through their respective local grid intervals, while rejected stretches reduce the measure rather than being filled in by the span between the smallest and largest accepted values. The calculation uses the grid indicators without interpolating critical-value crossings. It lies between zero and four and equals four when all 41 grid points are accepted. It is a finite-domain numerical summary, not an unrestricted confidence-region length. The Monte Carlo mean of F_r is the all-grid-points acceptance rate, which records how often every tested candidate in G_0 is accepted; it does not imply acceptance of the real line or global unboundedness.

The grid supports point-profile evaluation, the accepted-set measure, and the all-grid-points indicator. Coverage and power in Section 4.4 instead evaluate W_N directly at exact true or false candidate values and remain independent of membership in G_0.

4.6 Post-Main Numerical Diagnostics

After the main Monte Carlo design had been frozen and completed, a numerical-domain diagnostic examined whether confidence-set informativeness within the primary domain A_0 was sensitive to its finite endpoints. A finite grid cannot reveal what occurs beyond those endpoints, so acceptance near or at a boundary may cause the measured accepted set to reflect numerical truncation. The investigated domains were

$$A_0 = [-1, 3], \qquad A_1 = [-3, 5], \qquad A_2 = [-5, 7]. \tag{60}$$

The diagnostic used $n = 1000$, $R = 100$ diagnostic replications and primitive samples, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, and Oracle-GMM, Full-GMM, and DML-IVQR-BC. It used $\kappa \in \{1, 0.25, 0.10\}$, a subset selected to represent different levels of instrument relevance. Stage 1 evaluated $W_N(a)$ on the 81-point grid for A_1 with spacing 0.1; its 41-point A_0 summary came from the same profile. Thus, each replication held the primitive sample fixed across domains.

Stage 2 extended the investigated domain to A_2. It reloaded the 100 saved Stage 1 primitive samples, retained the 81 profile values on A_1, and evaluated only the 40 added tail points in $[-5, -3.1] \cup [5.1, 7]$. The resulting A_2 profile contained 121 points at the unchanged spacing of 0.1. Comparisons across domains therefore reflect additional candidate values rather than different Monte Carlo samples. For each domain, the implementation records the grid-based accepted-set measure, the accepted-grid share and count, acceptance at the left, right, either, and both boundaries, and whether all grid points are accepted. It also computes the change in accepted-set measure from A_0 to A_1 and from A_1 to A_2.

These diagnostics distinguish an accepted set numerically localized within a domain from one whose relevant portions continue beyond it. If acceptance reaches a numerical boundary, the warranted conclusion is only that the accepted set is not bounded within the domain examined. It does not establish that the theoretical confidence region is globally unbounded, that the entire real line is accepted, or that identification is completely absent. The diagnostic ended after Stage 2 at A_2; no Stage 3 or open-ended expansion was used. This keeps the exercise bounded rather than repeatedly enlarging the domain until a particular pattern emerges. Main Monte Carlo summaries continue to use A_0. The expansions diagnose the numerical interpretation of accepted-set informativeness, while coverage and power remain direct evaluations at their target values under Section 4.4. Substantive diagnostic findings are reported in Chapter 5.

5 Monte Carlo Results

Table 1. Instrument-strength calibration

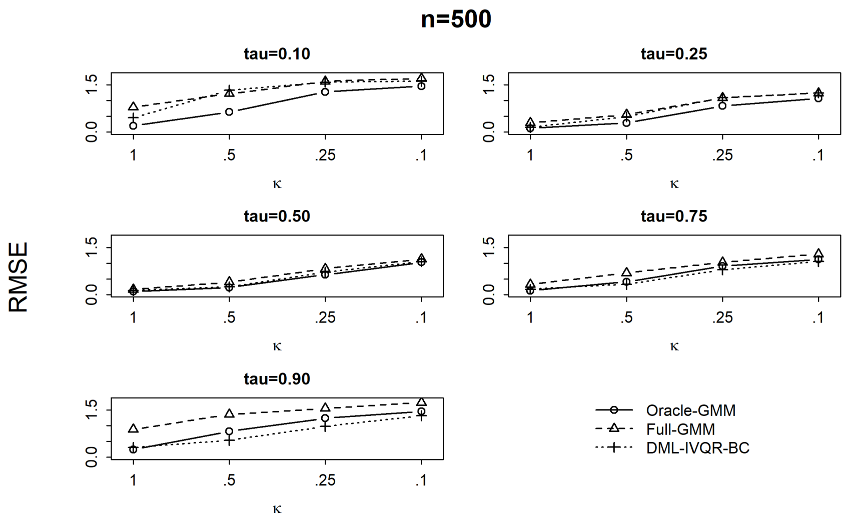
n	κ	**Median** F	F**: p10–p90**	**Median partial** R^2	**Partial** R^2**: p10–p90**
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

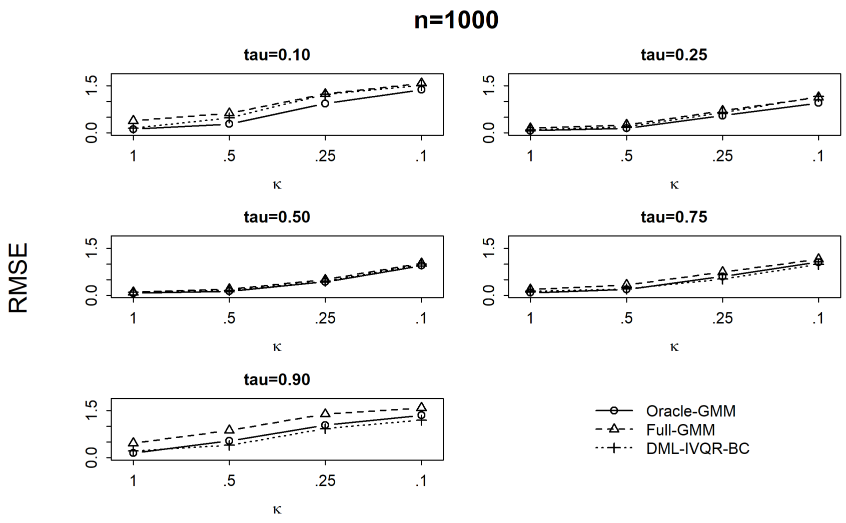
Table 2. Point-estimation performance at the median quantile, $n = 1000$

κ	**Estimator**	**Bias**	**MAE**	**RMSE**
1.00	Oracle-GMM	0.007	0.047	0.072
1.00	Full-GMM	0.007	0.072	0.100
1.00	DML-IVQR-BC	0.070	0.079	0.104
0.50	Oracle-GMM	0.015	0.093	0.134
0.50	Full-GMM	0.014	0.151	0.199
0.50	DML-IVQR-BC	0.059	0.119	0.160
0.25	Oracle-GMM	0.045	0.281	0.431
0.25	Full-GMM	0.029	0.372	0.500
0.25	DML-IVQR-BC	-0.003	0.306	0.448
0.10	Oracle-GMM	-0.044	0.735	0.945
0.10	Full-GMM	-0.012	0.840	1.018
0.10	DML-IVQR-BC	-0.195	0.779	0.991

Note: Each cell uses 500 Monte Carlo replications. The frozen sign convention is Bias = $\alpha_0 - \widehat{\alpha}$.



(a) $n = 500$



(b) $n = 1000$

Figure 1. RMSE across instrument-strength designs

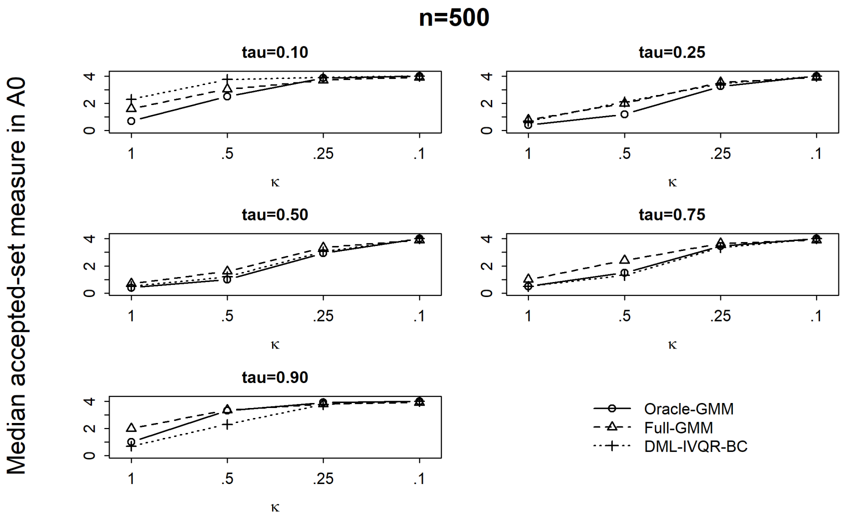


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 500$

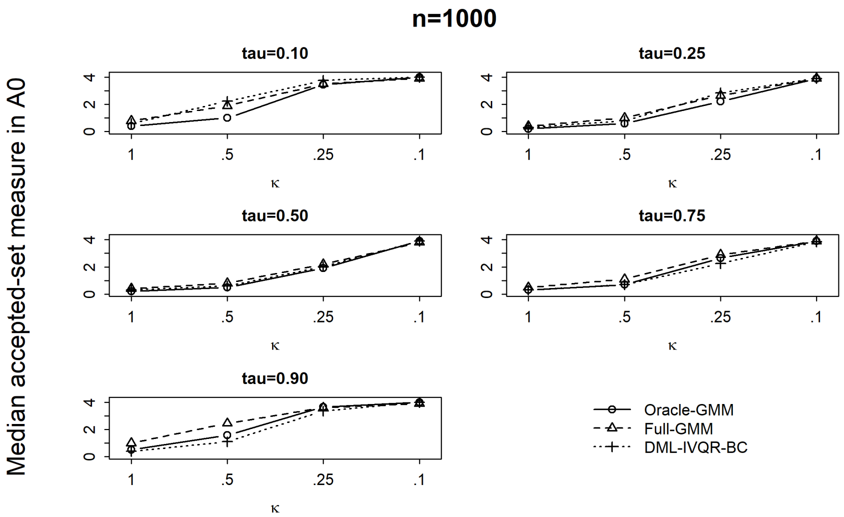


Figure 3. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 1000$

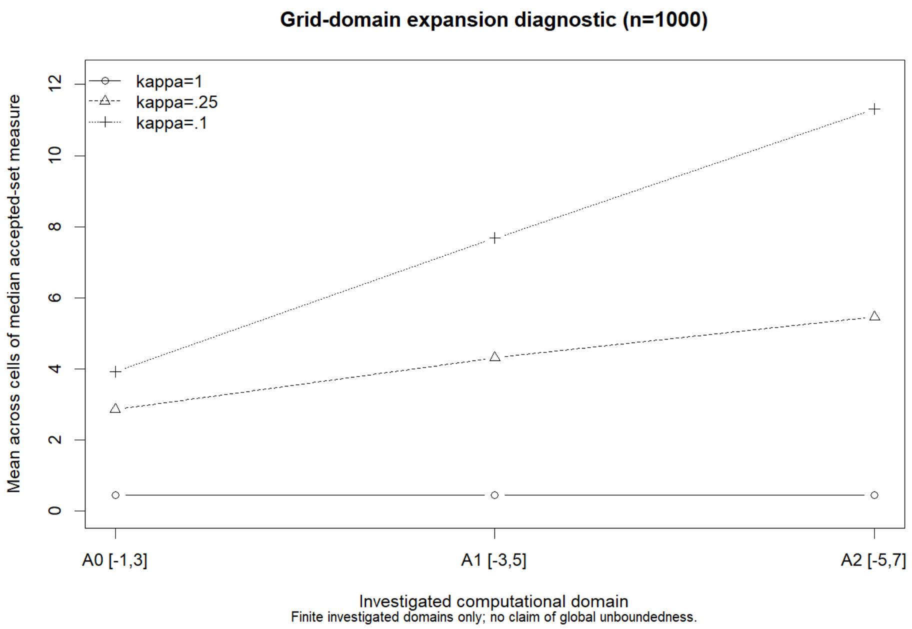


Figure 4. Sensitivity of the accepted set to expansion of the computational domain

Note: The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

5.1 Instrument-Strength Calibration

5.2 Point-Estimation Performance

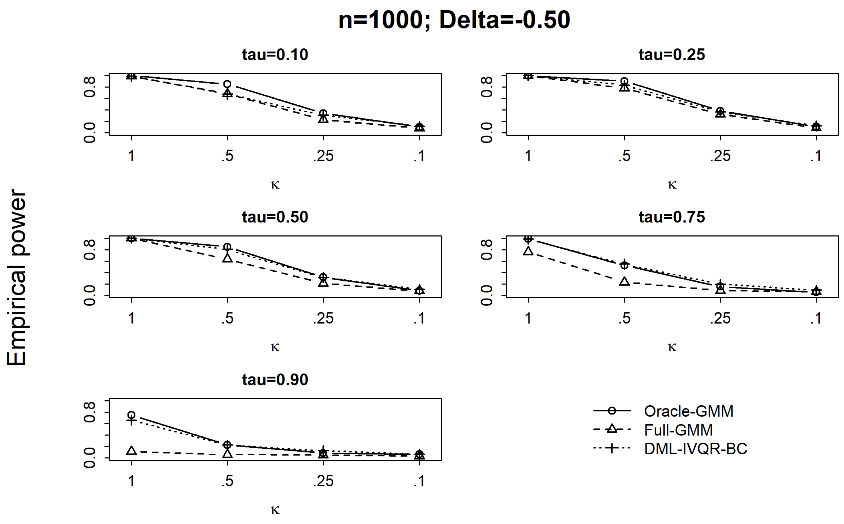
5.3 Confidence-Region Informativeness

5.4 Power

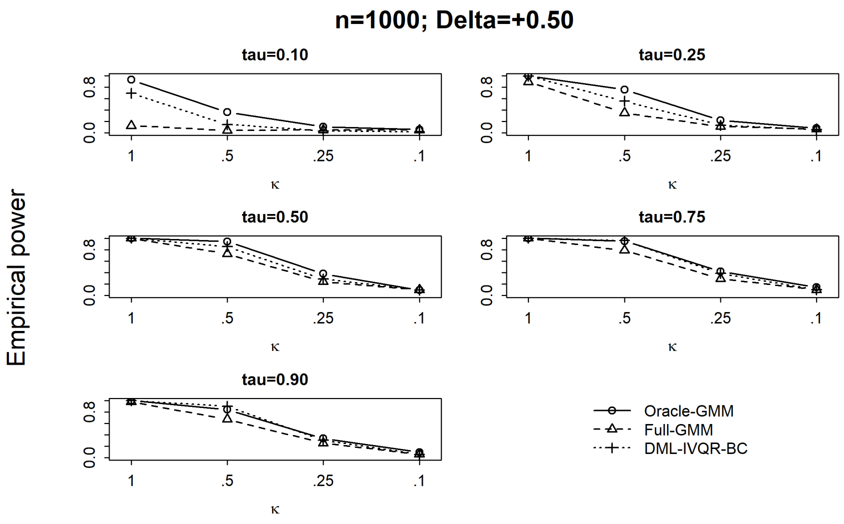
5.5 Coverage and Size

5.6 Heterogeneity Across Quantiles

5.7 Synthesis Across Estimators



(a) $\Delta = -0.50$



(b) $\Delta = +0.50$

Figure 5. Rejection probability under moderate false alternatives, $n = 1000$

Table 3. Finite-sample coverage under the strongest instrument design

τ	**Estimator**	**Coverage**	**Size**	**MCSE**	$q_{.95}(W_{\text{true}})$
0.10	Oracle-GMM	0.860	0.140	0.016	9.723
0.10	Full-GMM	0.502	0.498	0.022	21.090
0.10	DML-IVQR-BC	0.968	0.032	0.008	5.256
0.25	Oracle-GMM	0.874	0.126	0.015	8.300
0.25	Full-GMM	0.790	0.210	0.018	10.940
0.25	DML-IVQR-BC	0.894	0.106	0.014	7.811
0.50	Oracle-GMM	0.872	0.128	0.015	8.511
0.50	Full-GMM	0.896	0.104	0.014	7.524
0.50	DML-IVQR-BC	0.776	0.224	0.019	11.157
0.75	Oracle-GMM	0.860	0.140	0.016	8.763
0.75	Full-GMM	0.790	0.210	0.018	12.489
0.75	DML-IVQR-BC	0.632	0.368	0.022	16.034
0.90	Oracle-GMM	0.904	0.096	0.013	7.395
0.90	Full-GMM	0.530	0.470	0.022	20.072
0.90	DML-IVQR-BC	0.498	0.502	0.022	18.682

Note: Results use $n = 1000$, $\kappa = 1$, and 500 Monte Carlo replications. The unchanged 95% reference cutoff is $q_{\chi^2_2}(.95) = 5.991$.

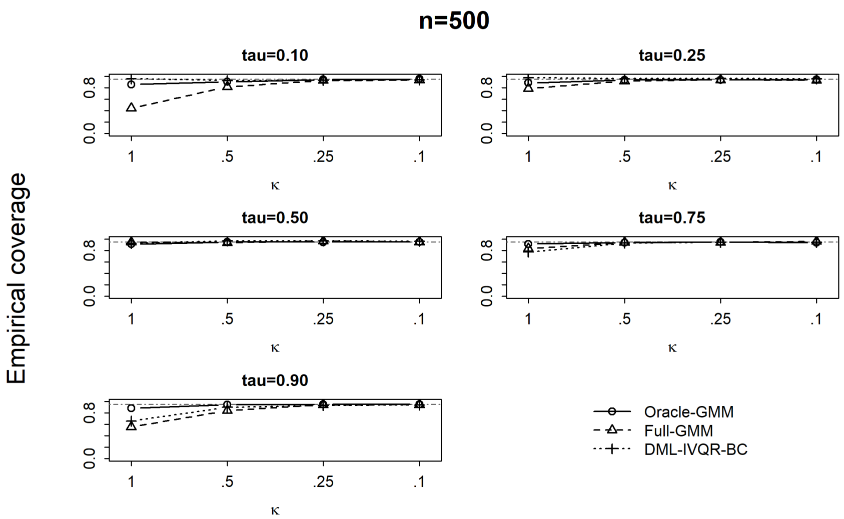


Figure 6. Empirical coverage across instrument-strength designs, $n = 500$

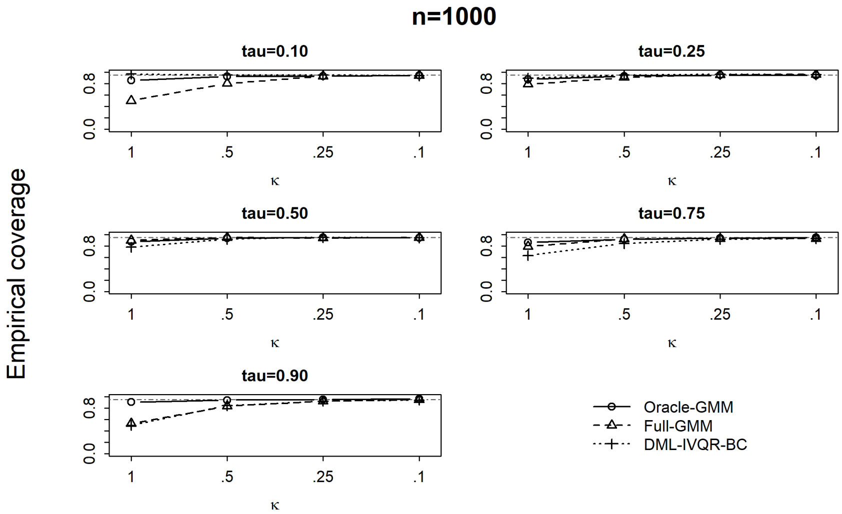


Figure 7. Empirical coverage across instrument-strength designs, $n = 1000$

6 Discussion

6.1 Main Findings and Weak Identification

6.2 Inferential Validity versus Informativeness

6.3 What DML Can and Cannot Address

6.4 Limitations

7 Conclusion

References

Chen, Jau-er, Chien-Hsun Huang, and Jia-Jyun Tien. 2021. "Debiased/Double Machine Learning for Instrumental Variable Quantile Regressions." *Econometrics* 9 (2): 15. https://doi.org/10.3390/econometrics9020015. [3–5, 7, 8, 10, 12, 14]

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, et al. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–C68. https://doi.org/10.1111/ectj.12097. [7, 10]

Chernozhukov, Victor, and Christian Hansen. 2005. "An IV Model of Quantile Treatment Effects." *Econometrica* 73 (1): 245–261. https://doi.org/10.1111/j.1468-0262.2005.00570.x. [2]

Chernozhukov, Victor, and Christian Hansen. 2008. "Instrumental Variable Quantile Regression: A Robust Inference Approach." *Journal of Econometrics* 142 (1): 379–398. https://doi.org/10.1016/j.jeconom.2007.06.005. [2, 13]

Chernozhukov, Victor, Christian Hansen, and Michael Jansson. 2009. "Finite Sample Inference for Quantile Regression Models." *Journal of Econometrics* 152 (2): 93–103. https://doi.org/10.1016/j.jeconom.2009.01.004. [13]

Appendix A Full Monte Carlo Results

Table A.1. Full point-estimation performance

n	τ	κ	Estimator	Bias	MAE	RMSE
500	0.10	1.00	Oracle-GMM	-0.115	0.155	0.200
500	0.10	1.00	Full-GMM	-0.684	0.693	0.789
500	0.10	1.00	DML-IVQR-BC	-0.204	0.296	0.453
500	0.10	0.50	Oracle-GMM	-0.285	0.394	0.642
500	0.10	0.50	Full-GMM	-0.901	0.960	1.219
500	0.10	0.50	DML-IVQR-BC	-0.827	0.926	1.335
500	0.10	0.25	Oracle-GMM	-0.705	0.895	1.276
500	0.10	0.25	Full-GMM	-1.155	1.283	1.612
500	0.10	0.25	DML-IVQR-BC	-1.059	1.207	1.585
500	0.10	0.10	Oracle-GMM	-0.847	1.097	1.457
500	0.10	0.10	Full-GMM	-1.272	1.396	1.707
500	0.10	0.10	DML-IVQR-BC	-1.065	1.258	1.618
500	0.25	1.00	Oracle-GMM	-0.048	0.097	0.123
500	0.25	1.00	Full-GMM	-0.218	0.246	0.295
500	0.25	1.00	DML-IVQR-BC	0.006	0.118	0.150
500	0.25	0.50	Oracle-GMM	-0.083	0.213	0.294
500	0.25	0.50	Full-GMM	-0.272	0.423	0.553
500	0.25	0.50	DML-IVQR-BC	-0.156	0.311	0.486
500	0.25	0.25	Oracle-GMM	-0.264	0.579	0.831
500	0.25	0.25	Full-GMM	-0.502	0.844	1.088
500	0.25	0.25	DML-IVQR-BC	-0.576	0.801	1.092
500	0.25	0.10	Oracle-GMM	-0.312	0.833	1.071
500	0.25	0.10	Full-GMM	-0.549	1.006	1.241
500	0.25	0.10	DML-IVQR-BC	-0.601	0.987	1.245
500	0.50	1.00	Oracle-GMM	-0.010	0.075	0.102
500	0.50	1.00	Full-GMM	0.009	0.131	0.169
500	0.50	1.00	DML-IVQR-BC	0.062	0.100	0.131
500	0.50	0.50	Oracle-GMM	-0.007	0.164	0.228
500	0.50	0.50	Full-GMM	0.011	0.303	0.400
500	0.50	0.50	DML-IVQR-BC	0.010	0.183	0.247
500	0.50	0.25	Oracle-GMM	-0.033	0.442	0.634
500	0.50	0.25	Full-GMM	0.008	0.640	0.827
500	0.50	0.25	DML-IVQR-BC	-0.199	0.499	0.715
500	0.50	0.10	Oracle-GMM	-0.060	0.818	1.027
500	0.50	0.10	Full-GMM	0.036	0.949	1.129
500	0.50	0.10	DML-IVQR-BC	-0.268	0.848	1.046

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TableA.1. Full point-estimation performance (Continued)

n	τ	κ	**Estimator**	**Bias**	**MAE**	**RMSE**
500	0.75	1.00	Oracle-GMM	0.034	0.097	0.126
500	0.75	1.00	Full-GMM	0.231	0.266	0.327
500	0.75	1.00	DML-IVQR-BC	0.125	0.144	0.175
500	0.75	0.50	Oracle-GMM	0.117	0.251	0.410
500	0.75	0.50	Full-GMM	0.325	0.499	0.686
500	0.75	0.50	DML-IVQR-BC	0.118	0.238	0.329
500	0.75	0.25	Oracle-GMM	0.300	0.613	0.903
500	0.75	0.25	Full-GMM	0.417	0.798	1.030
500	0.75	0.25	DML-IVQR-BC	0.106	0.562	0.793
500	0.75	0.10	Oracle-GMM	0.260	0.868	1.123
500	0.75	0.10	Full-GMM	0.488	1.048	1.288
500	0.75	0.10	DML-IVQR-BC	0.159	0.828	1.055
500	0.90	1.00	Oracle-GMM	0.111	0.179	0.246
500	0.90	1.00	Full-GMM	0.736	0.751	0.880
500	0.90	1.00	DML-IVQR-BC	0.244	0.254	0.306
500	0.90	0.50	Oracle-GMM	0.348	0.510	0.822
500	0.90	0.50	Full-GMM	1.005	1.087	1.362
500	0.90	0.50	DML-IVQR-BC	0.286	0.373	0.539
500	0.90	0.25	Oracle-GMM	0.594	0.842	1.237
500	0.90	0.25	Full-GMM	1.071	1.230	1.551
500	0.90	0.25	DML-IVQR-BC	0.378	0.663	0.975
500	0.90	0.10	Oracle-GMM	0.812	1.097	1.456
500	0.90	0.10	Full-GMM	1.266	1.416	1.730
500	0.90	0.10	DML-IVQR-BC	0.648	0.968	1.322
1000	0.10	1.00	Oracle-GMM	-0.054	0.093	0.121
1000	0.10	1.00	Full-GMM	-0.329	0.337	0.387
1000	0.10	1.00	DML-IVQR-BC	-0.005	0.117	0.150
1000	0.10	0.50	Oracle-GMM	-0.101	0.194	0.287
1000	0.10	0.50	Full-GMM	-0.407	0.465	0.630
1000	0.10	0.50	DML-IVQR-BC	-0.159	0.306	0.482
1000	0.10	0.25	Oracle-GMM	-0.384	0.566	0.934
1000	0.10	0.25	Full-GMM	-0.773	0.914	1.242
1000	0.10	0.25	DML-IVQR-BC	-0.681	0.844	1.229
1000	0.10	0.10	Oracle-GMM	-0.747	0.975	1.376
1000	0.10	0.10	Full-GMM	-1.074	1.251	1.582
1000	0.10	0.10	DML-IVQR-BC	-0.932	1.146	1.514
1000	0.25	1.00	Oracle-GMM	-0.018	0.063	0.077
1000	0.25	1.00	Full-GMM	-0.098	0.126	0.153
1000	0.25	1.00	DML-IVQR-BC	0.044	0.078	0.097
1000	0.25	0.50	Oracle-GMM	-0.029	0.113	0.147

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TableA.1. Full point-estimation performance (Continued)

n	τ	κ	Estimator	Bias	MAE	RMSE
1000	0.25	0.50	Full-GMM	-0.114	0.201	0.259
1000	0.25	0.50	DML-IVQR-BC	-0.000	0.145	0.198
1000	0.25	0.25	Oracle-GMM	-0.131	0.332	0.550
1000	0.25	0.25	Full-GMM	-0.232	0.486	0.706
1000	0.25	0.25	DML-IVQR-BC	-0.199	0.405	0.647
1000	0.25	0.10	Oracle-GMM	-0.235	0.696	0.954
1000	0.25	0.10	Full-GMM	-0.410	0.878	1.136
1000	0.25	0.10	DML-IVQR-BC	-0.490	0.862	1.157
1000	0.50	1.00	Oracle-GMM	0.007	0.047	0.072
1000	0.50	1.00	Full-GMM	0.007	0.072	0.100
1000	0.50	1.00	DML-IVQR-BC	0.070	0.079	0.104
1000	0.50	0.50	Oracle-GMM	0.015	0.093	0.134
1000	0.50	0.50	Full-GMM	0.014	0.151	0.199
1000	0.50	0.50	DML-IVQR-BC	0.059	0.119	0.160
1000	0.50	0.25	Oracle-GMM	0.045	0.281	0.431
1000	0.50	0.25	Full-GMM	0.029	0.372	0.500
1000	0.50	0.25	DML-IVQR-BC	-0.003	0.306	0.448
1000	0.50	0.10	Oracle-GMM	-0.044	0.735	0.945
1000	0.50	0.10	Full-GMM	-0.012	0.840	1.018
1000	0.50	0.10	DML-IVQR-BC	-0.195	0.779	0.991
1000	0.75	1.00	Oracle-GMM	0.025	0.068	0.085
1000	0.75	1.00	Full-GMM	0.126	0.151	0.186
1000	0.75	1.00	DML-IVQR-BC	0.105	0.113	0.133
1000	0.75	0.50	Oracle-GMM	0.055	0.134	0.189
1000	0.75	0.50	Full-GMM	0.175	0.244	0.332
1000	0.75	0.50	DML-IVQR-BC	0.111	0.149	0.203
1000	0.75	0.25	Oracle-GMM	0.186	0.381	0.604
1000	0.75	0.25	Full-GMM	0.304	0.517	0.745
1000	0.75	0.25	DML-IVQR-BC	0.132	0.336	0.516
1000	0.75	0.10	Oracle-GMM	0.256	0.788	1.058
1000	0.75	0.10	Full-GMM	0.422	0.907	1.147
1000	0.75	0.10	DML-IVQR-BC	0.136	0.753	0.982
1000	0.90	1.00	Oracle-GMM	0.072	0.110	0.148
1000	0.90	1.00	Full-GMM	0.387	0.397	0.465
1000	0.90	1.00	DML-IVQR-BC	0.183	0.186	0.215
1000	0.90	0.50	Oracle-GMM	0.211	0.295	0.540
1000	0.90	0.50	Full-GMM	0.584	0.629	0.871
1000	0.90	0.50	DML-IVQR-BC	0.226	0.260	0.396
1000	0.90	0.25	Oracle-GMM	0.491	0.658	1.038
1000	0.90	0.25	Full-GMM	0.901	1.058	1.389

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TableA.1. Full point-estimation performance (Continued)

n	τ	κ	Estimator	Bias	MAE	RMSE
1000	0.90	0.25	DML-IVQR-BC	0.417	0.580	0.927
1000	0.90	0.10	Oracle-GMM	0.653	0.948	1.350
1000	0.90	0.10	Full-GMM	1.050	1.236	1.583
1000	0.90	0.10	DML-IVQR-BC	0.519	0.831	1.191

Note: All 120 Monte Carlo design cells. Bias follows $\alpha_0 - \widehat{\alpha}$; each cell uses 500 replications.

Appendix B Detailed Power Results

Table B.1. Full power results

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.10	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.10	1.00	Oracle-GMM	-0.50	0.968	0.032	500	0
500	0.10	1.00	Oracle-GMM	-0.25	0.694	0.306	500	0
500	0.10	1.00	Oracle-GMM	0.25	0.120	0.880	500	0
500	0.10	1.00	Oracle-GMM	0.50	0.482	0.518	500	0
500	0.10	1.00	Oracle-GMM	1.00	0.930	0.070	500	0
500	0.10	1.00	Full-GMM	-1.00	0.970	0.030	500	0
500	0.10	1.00	Full-GMM	-0.50	0.898	0.102	500	0
500	0.10	1.00	Full-GMM	-0.25	0.770	0.230	500	0
500	0.10	1.00	Full-GMM	0.25	0.248	0.752	500	0
500	0.10	1.00	Full-GMM	0.50	0.050	0.950	500	0
500	0.10	1.00	Full-GMM	1.00	0.134	0.866	500	0
500	0.10	1.00	DML-IVQR-BC	-1.00	0.998	0.002	500	0
500	0.10	1.00	DML-IVQR-BC	-0.50	0.744	0.256	500	0
500	0.10	1.00	DML-IVQR-BC	-0.25	0.262	0.738	500	0
500	0.10	1.00	DML-IVQR-BC	0.25	0.028	0.972	500	0
500	0.10	1.00	DML-IVQR-BC	0.50	0.058	0.942	500	0
500	0.10	1.00	DML-IVQR-BC	1.00	0.184	0.816	500	0
500	0.10	0.50	Oracle-GMM	-1.00	0.794	0.206	500	0
500	0.10	0.50	Oracle-GMM	-0.50	0.532	0.468	500	0
500	0.10	0.50	Oracle-GMM	-0.25	0.292	0.708	500	0
500	0.10	0.50	Oracle-GMM	0.25	0.090	0.910	500	0
500	0.10	0.50	Oracle-GMM	0.50	0.144	0.856	500	0
500	0.10	0.50	Oracle-GMM	1.00	0.308	0.692	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.10	0.50	Full-GMM	-1.00	0.558	0.442	500	0
500	0.10	0.50	Full-GMM	-0.50	0.438	0.562	500	0
500	0.10	0.50	Full-GMM	-0.25	0.310	0.690	500	0
500	0.10	0.50	Full-GMM	0.25	0.108	0.892	500	0
500	0.10	0.50	Full-GMM	0.50	0.068	0.932	500	0
500	0.10	0.50	Full-GMM	1.00	0.082	0.918	500	0
500	0.10	0.50	DML-IVQR-BC	-1.00	0.834	0.166	500	0
500	0.10	0.50	DML-IVQR-BC	-0.50	0.406	0.594	500	0
500	0.10	0.50	DML-IVQR-BC	-0.25	0.174	0.826	500	0
500	0.10	0.50	DML-IVQR-BC	0.25	0.038	0.962	500	0
500	0.10	0.50	DML-IVQR-BC	0.50	0.042	0.958	500	0
500	0.10	0.50	DML-IVQR-BC	1.00	0.032	0.968	500	0
500	0.10	0.25	Oracle-GMM	-1.00	0.246	0.754	500	0
500	0.10	0.25	Oracle-GMM	-0.50	0.188	0.812	500	0
500	0.10	0.25	Oracle-GMM	-0.25	0.112	0.888	500	0
500	0.10	0.25	Oracle-GMM	0.25	0.050	0.950	500	0
500	0.10	0.25	Oracle-GMM	0.50	0.074	0.926	500	0
500	0.10	0.25	Oracle-GMM	1.00	0.104	0.896	500	0
500	0.10	0.25	Full-GMM	-1.00	0.168	0.832	500	0
500	0.10	0.25	Full-GMM	-0.50	0.134	0.866	500	0
500	0.10	0.25	Full-GMM	-0.25	0.106	0.894	500	0
500	0.10	0.25	Full-GMM	0.25	0.052	0.948	500	0
500	0.10	0.25	Full-GMM	0.50	0.050	0.950	500	0
500	0.10	0.25	Full-GMM	1.00	0.026	0.974	500	0
500	0.10	0.25	DML-IVQR-BC	-1.00	0.432	0.568	500	0
500	0.10	0.25	DML-IVQR-BC	-0.50	0.224	0.776	500	0
500	0.10	0.25	DML-IVQR-BC	-0.25	0.116	0.884	500	0
500	0.10	0.25	DML-IVQR-BC	0.25	0.046	0.954	500	0
500	0.10	0.25	DML-IVQR-BC	0.50	0.036	0.964	500	0
500	0.10	0.25	DML-IVQR-BC	1.00	0.036	0.964	500	0
500	0.10	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.10	0.10	Oracle-GMM	-0.50	0.090	0.910	500	0
500	0.10	0.10	Oracle-GMM	-0.25	0.062	0.938	500	0
500	0.10	0.10	Oracle-GMM	0.25	0.052	0.948	500	0
500	0.10	0.10	Oracle-GMM	0.50	0.070	0.930	500	0
500	0.10	0.10	Oracle-GMM	1.00	0.056	0.944	500	0
500	0.10	0.10	Full-GMM	-1.00	0.056	0.944	500	0
500	0.10	0.10	Full-GMM	-0.50	0.052	0.948	500	0
500	0.10	0.10	Full-GMM	-0.25	0.072	0.928	500	0
500	0.10	0.10	Full-GMM	0.25	0.070	0.930	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.10	0.10	Full-GMM	0.50	0.058	0.942	500	0
500	0.10	0.10	Full-GMM	1.00	0.038	0.962	500	0
500	0.10	0.10	DML-IVQR-BC	-1.00	0.194	0.806	500	0
500	0.10	0.10	DML-IVQR-BC	-0.50	0.136	0.864	500	0
500	0.10	0.10	DML-IVQR-BC	-0.25	0.100	0.900	500	0
500	0.10	0.10	DML-IVQR-BC	0.25	0.058	0.942	500	0
500	0.10	0.10	DML-IVQR-BC	0.50	0.062	0.938	500	0
500	0.10	0.10	DML-IVQR-BC	1.00	0.054	0.946	500	0
500	0.25	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.25	1.00	Oracle-GMM	-0.50	0.994	0.006	500	0
500	0.25	1.00	Oracle-GMM	-0.25	0.818	0.182	500	0
500	0.25	1.00	Oracle-GMM	0.25	0.450	0.550	500	0
500	0.25	1.00	Oracle-GMM	0.50	0.930	0.070	500	0
500	0.25	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.25	1.00	Full-GMM	-1.00	1.000	0.000	500	0
500	0.25	1.00	Full-GMM	-0.50	0.928	0.072	500	0
500	0.25	1.00	Full-GMM	-0.25	0.692	0.308	500	0
500	0.25	1.00	Full-GMM	0.25	0.052	0.948	500	0
500	0.25	1.00	Full-GMM	0.50	0.242	0.758	500	0
500	0.25	1.00	Full-GMM	1.00	0.908	0.092	500	0
500	0.25	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
500	0.25	1.00	DML-IVQR-BC	-0.50	0.940	0.060	500	0
500	0.25	1.00	DML-IVQR-BC	-0.25	0.402	0.598	500	0
500	0.25	1.00	DML-IVQR-BC	0.25	0.248	0.752	500	0
500	0.25	1.00	DML-IVQR-BC	0.50	0.668	0.332	500	0
500	0.25	1.00	DML-IVQR-BC	1.00	0.980	0.020	500	0
500	0.25	0.50	Oracle-GMM	-1.00	0.906	0.094	500	0
500	0.25	0.50	Oracle-GMM	-0.50	0.618	0.382	500	0
500	0.25	0.50	Oracle-GMM	-0.25	0.330	0.670	500	0
500	0.25	0.50	Oracle-GMM	0.25	0.138	0.862	500	0
500	0.25	0.50	Oracle-GMM	0.50	0.330	0.670	500	0
500	0.25	0.50	Oracle-GMM	1.00	0.654	0.346	500	0
500	0.25	0.50	Full-GMM	-1.00	0.762	0.238	500	0
500	0.25	0.50	Full-GMM	-0.50	0.448	0.552	500	0
500	0.25	0.50	Full-GMM	-0.25	0.242	0.758	500	0
500	0.25	0.50	Full-GMM	0.25	0.054	0.946	500	0
500	0.25	0.50	Full-GMM	0.50	0.092	0.908	500	0
500	0.25	0.50	Full-GMM	1.00	0.308	0.692	500	0
500	0.25	0.50	DML-IVQR-BC	-1.00	0.930	0.070	500	0
500	0.25	0.50	DML-IVQR-BC	-0.50	0.530	0.470	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.25	0.50	DML-IVQR-BC	-0.25	0.170	0.830	500	0
500	0.25	0.50	DML-IVQR-BC	0.25	0.042	0.958	500	0
500	0.25	0.50	DML-IVQR-BC	0.50	0.144	0.856	500	0
500	0.25	0.50	DML-IVQR-BC	1.00	0.316	0.684	500	0
500	0.25	0.25	Oracle-GMM	-1.00	0.370	0.630	500	0
500	0.25	0.25	Oracle-GMM	-0.50	0.224	0.776	500	0
500	0.25	0.25	Oracle-GMM	-0.25	0.106	0.894	500	0
500	0.25	0.25	Oracle-GMM	0.25	0.072	0.928	500	0
500	0.25	0.25	Oracle-GMM	0.50	0.100	0.900	500	0
500	0.25	0.25	Oracle-GMM	1.00	0.178	0.822	500	0
500	0.25	0.25	Full-GMM	-1.00	0.292	0.708	500	0
500	0.25	0.25	Full-GMM	-0.50	0.184	0.816	500	0
500	0.25	0.25	Full-GMM	-0.25	0.120	0.880	500	0
500	0.25	0.25	Full-GMM	0.25	0.054	0.946	500	0
500	0.25	0.25	Full-GMM	0.50	0.070	0.930	500	0
500	0.25	0.25	Full-GMM	1.00	0.122	0.878	500	0
500	0.25	0.25	DML-IVQR-BC	-1.00	0.456	0.544	500	0
500	0.25	0.25	DML-IVQR-BC	-0.50	0.214	0.786	500	0
500	0.25	0.25	DML-IVQR-BC	-0.25	0.098	0.902	500	0
500	0.25	0.25	DML-IVQR-BC	0.25	0.030	0.970	500	0
500	0.25	0.25	DML-IVQR-BC	0.50	0.052	0.948	500	0
500	0.25	0.25	DML-IVQR-BC	1.00	0.064	0.936	500	0
500	0.25	0.10	Oracle-GMM	-1.00	0.104	0.896	500	0
500	0.25	0.10	Oracle-GMM	-0.50	0.078	0.922	500	0
500	0.25	0.10	Oracle-GMM	-0.25	0.070	0.930	500	0
500	0.25	0.10	Oracle-GMM	0.25	0.066	0.934	500	0
500	0.25	0.10	Oracle-GMM	0.50	0.050	0.950	500	0
500	0.25	0.10	Oracle-GMM	1.00	0.064	0.936	500	0
500	0.25	0.10	Full-GMM	-1.00	0.108	0.892	500	0
500	0.25	0.10	Full-GMM	-0.50	0.068	0.932	500	0
500	0.25	0.10	Full-GMM	-0.25	0.078	0.922	500	0
500	0.25	0.10	Full-GMM	0.25	0.044	0.956	500	0
500	0.25	0.10	Full-GMM	0.50	0.050	0.950	500	0
500	0.25	0.10	Full-GMM	1.00	0.072	0.928	500	0
500	0.25	0.10	DML-IVQR-BC	-1.00	0.168	0.832	500	0
500	0.25	0.10	DML-IVQR-BC	-0.50	0.114	0.886	500	0
500	0.25	0.10	DML-IVQR-BC	-0.25	0.072	0.928	500	0
500	0.25	0.10	DML-IVQR-BC	0.25	0.038	0.962	500	0
500	0.25	0.10	DML-IVQR-BC	0.50	0.034	0.966	500	0
500	0.25	0.10	DML-IVQR-BC	1.00	0.044	0.956	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.50	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
500	0.50	1.00	Oracle-GMM	-0.50	0.990	0.010	500	0
500	0.50	1.00	Oracle-GMM	-0.25	0.730	0.270	500	0
500	0.50	1.00	Oracle-GMM	0.25	0.734	0.266	500	0
500	0.50	1.00	Oracle-GMM	0.50	0.994	0.006	500	0
500	0.50	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.50	1.00	Full-GMM	-1.00	1.000	0.000	500	0
500	0.50	1.00	Full-GMM	-0.50	0.790	0.210	500	0
500	0.50	1.00	Full-GMM	-0.25	0.312	0.688	500	0
500	0.50	1.00	Full-GMM	0.25	0.324	0.676	500	0
500	0.50	1.00	Full-GMM	0.50	0.854	0.146	500	0
500	0.50	1.00	Full-GMM	1.00	1.000	0.000	500	0
500	0.50	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
500	0.50	1.00	DML-IVQR-BC	-0.50	0.962	0.038	500	0
500	0.50	1.00	DML-IVQR-BC	-0.25	0.398	0.602	500	0
500	0.50	1.00	DML-IVQR-BC	0.25	0.716	0.284	500	0
500	0.50	1.00	DML-IVQR-BC	0.50	0.984	0.016	500	0
500	0.50	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.50	0.50	Oracle-GMM	-1.00	0.838	0.162	500	0
500	0.50	0.50	Oracle-GMM	-0.50	0.552	0.448	500	0
500	0.50	0.50	Oracle-GMM	-0.25	0.230	0.770	500	0
500	0.50	0.50	Oracle-GMM	0.25	0.220	0.780	500	0
500	0.50	0.50	Oracle-GMM	0.50	0.560	0.440	500	0
500	0.50	0.50	Oracle-GMM	1.00	0.924	0.076	500	0
500	0.50	0.50	Full-GMM	-1.00	0.630	0.370	500	0
500	0.50	0.50	Full-GMM	-0.50	0.260	0.740	500	0
500	0.50	0.50	Full-GMM	-0.25	0.122	0.878	500	0
500	0.50	0.50	Full-GMM	0.25	0.116	0.884	500	0
500	0.50	0.50	Full-GMM	0.50	0.276	0.724	500	0
500	0.50	0.50	Full-GMM	1.00	0.704	0.296	500	0
500	0.50	0.50	DML-IVQR-BC	-1.00	0.894	0.106	500	0
500	0.50	0.50	DML-IVQR-BC	-0.50	0.466	0.534	500	0
500	0.50	0.50	DML-IVQR-BC	-0.25	0.156	0.844	500	0
500	0.50	0.50	DML-IVQR-BC	0.25	0.144	0.856	500	0
500	0.50	0.50	DML-IVQR-BC	0.50	0.392	0.608	500	0
500	0.50	0.50	DML-IVQR-BC	1.00	0.754	0.246	500	0
500	0.50	0.25	Oracle-GMM	-1.00	0.294	0.706	500	0
500	0.50	0.25	Oracle-GMM	-0.50	0.144	0.856	500	0
500	0.50	0.25	Oracle-GMM	-0.25	0.096	0.904	500	0
500	0.50	0.25	Oracle-GMM	0.25	0.090	0.910	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.50	0.25	Oracle-GMM	0.50	0.164	0.836	500	0
500	0.50	0.25	Oracle-GMM	1.00	0.330	0.670	500	0
500	0.50	0.25	Full-GMM	-1.00	0.202	0.798	500	0
500	0.50	0.25	Full-GMM	-0.50	0.094	0.906	500	0
500	0.50	0.25	Full-GMM	-0.25	0.080	0.920	500	0
500	0.50	0.25	Full-GMM	0.25	0.052	0.948	500	0
500	0.50	0.25	Full-GMM	0.50	0.082	0.918	500	0
500	0.50	0.25	Full-GMM	1.00	0.216	0.784	500	0
500	0.50	0.25	DML-IVQR-BC	-1.00	0.354	0.646	500	0
500	0.50	0.25	DML-IVQR-BC	-0.50	0.168	0.832	500	0
500	0.50	0.25	DML-IVQR-BC	-0.25	0.094	0.906	500	0
500	0.50	0.25	DML-IVQR-BC	0.25	0.054	0.946	500	0
500	0.50	0.25	DML-IVQR-BC	0.50	0.072	0.928	500	0
500	0.50	0.25	DML-IVQR-BC	1.00	0.178	0.822	500	0
500	0.50	0.10	Oracle-GMM	-1.00	0.102	0.898	500	0
500	0.50	0.10	Oracle-GMM	-0.50	0.074	0.926	500	0
500	0.50	0.10	Oracle-GMM	-0.25	0.068	0.932	500	0
500	0.50	0.10	Oracle-GMM	0.25	0.066	0.934	500	0
500	0.50	0.10	Oracle-GMM	0.50	0.062	0.938	500	0
500	0.50	0.10	Oracle-GMM	1.00	0.062	0.938	500	0
500	0.50	0.10	Full-GMM	-1.00	0.098	0.902	500	0
500	0.50	0.10	Full-GMM	-0.50	0.088	0.912	500	0
500	0.50	0.10	Full-GMM	-0.25	0.064	0.936	500	0
500	0.50	0.10	Full-GMM	0.25	0.044	0.956	500	0
500	0.50	0.10	Full-GMM	0.50	0.060	0.940	500	0
500	0.50	0.10	Full-GMM	1.00	0.088	0.912	500	0
500	0.50	0.10	DML-IVQR-BC	-1.00	0.116	0.884	500	0
500	0.50	0.10	DML-IVQR-BC	-0.50	0.104	0.896	500	0
500	0.50	0.10	DML-IVQR-BC	-0.25	0.066	0.934	500	0
500	0.50	0.10	DML-IVQR-BC	0.25	0.032	0.968	500	0
500	0.50	0.10	DML-IVQR-BC	0.50	0.030	0.970	500	0
500	0.50	0.10	DML-IVQR-BC	1.00	0.056	0.944	500	0
500	0.75	1.00	Oracle-GMM	-1.00	0.990	0.010	500	0
500	0.75	1.00	Oracle-GMM	-0.50	0.832	0.168	500	0
500	0.75	1.00	Oracle-GMM	-0.25	0.366	0.634	500	0
500	0.75	1.00	Oracle-GMM	0.25	0.772	0.228	500	0
500	0.75	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
500	0.75	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.75	1.00	Full-GMM	-1.00	0.790	0.210	500	0
500	0.75	1.00	Full-GMM	-0.50	0.202	0.798	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.75	1.00	Full-GMM	-0.25	0.034	0.966	500	0
500	0.75	1.00	Full-GMM	0.25	0.608	0.392	500	0
500	0.75	1.00	Full-GMM	0.50	0.922	0.078	500	0
500	0.75	1.00	Full-GMM	1.00	1.000	0.000	500	0
500	0.75	1.00	DML-IVQR-BC	-1.00	0.998	0.002	500	0
500	0.75	1.00	DML-IVQR-BC	-0.50	0.754	0.246	500	0
500	0.75	1.00	DML-IVQR-BC	-0.25	0.160	0.840	500	0
500	0.75	1.00	DML-IVQR-BC	0.25	0.896	0.104	500	0
500	0.75	1.00	DML-IVQR-BC	0.50	0.998	0.002	500	0
500	0.75	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.75	0.50	Oracle-GMM	-1.00	0.510	0.490	500	0
500	0.75	0.50	Oracle-GMM	-0.50	0.252	0.748	500	0
500	0.75	0.50	Oracle-GMM	-0.25	0.124	0.876	500	0
500	0.75	0.50	Oracle-GMM	0.25	0.286	0.714	500	0
500	0.75	0.50	Oracle-GMM	0.50	0.646	0.354	500	0
500	0.75	0.50	Oracle-GMM	1.00	0.932	0.068	500	0
500	0.75	0.50	Full-GMM	-1.00	0.234	0.766	500	0
500	0.75	0.50	Full-GMM	-0.50	0.058	0.942	500	0
500	0.75	0.50	Full-GMM	-0.25	0.040	0.960	500	0
500	0.75	0.50	Full-GMM	0.25	0.200	0.800	500	0
500	0.75	0.50	Full-GMM	0.50	0.410	0.590	500	0
500	0.75	0.50	Full-GMM	1.00	0.740	0.260	500	0
500	0.75	0.50	DML-IVQR-BC	-1.00	0.614	0.386	500	0
500	0.75	0.50	DML-IVQR-BC	-0.50	0.284	0.716	500	0
500	0.75	0.50	DML-IVQR-BC	-0.25	0.088	0.912	500	0
500	0.75	0.50	DML-IVQR-BC	0.25	0.304	0.696	500	0
500	0.75	0.50	DML-IVQR-BC	0.50	0.574	0.426	500	0
500	0.75	0.50	DML-IVQR-BC	1.00	0.844	0.156	500	0
500	0.75	0.25	Oracle-GMM	-1.00	0.150	0.850	500	0
500	0.75	0.25	Oracle-GMM	-0.50	0.086	0.914	500	0
500	0.75	0.25	Oracle-GMM	-0.25	0.060	0.940	500	0
500	0.75	0.25	Oracle-GMM	0.25	0.112	0.888	500	0
500	0.75	0.25	Oracle-GMM	0.50	0.184	0.816	500	0
500	0.75	0.25	Oracle-GMM	1.00	0.358	0.642	500	0
500	0.75	0.25	Full-GMM	-1.00	0.092	0.908	500	0
500	0.75	0.25	Full-GMM	-0.50	0.036	0.964	500	0
500	0.75	0.25	Full-GMM	-0.25	0.040	0.960	500	0
500	0.75	0.25	Full-GMM	0.25	0.074	0.926	500	0
500	0.75	0.25	Full-GMM	0.50	0.106	0.894	500	0
500	0.75	0.25	Full-GMM	1.00	0.248	0.752	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
500	0.75	0.25	DML-IVQR-BC	-1.00	0.212	0.788	500	0
500	0.75	0.25	DML-IVQR-BC	-0.50	0.112	0.888	500	0
500	0.75	0.25	DML-IVQR-BC	-0.25	0.070	0.930	500	0
500	0.75	0.25	DML-IVQR-BC	0.25	0.084	0.916	500	0
500	0.75	0.25	DML-IVQR-BC	0.50	0.158	0.842	500	0
500	0.75	0.25	DML-IVQR-BC	1.00	0.230	0.770	500	0
500	0.75	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.75	0.10	Oracle-GMM	-0.50	0.056	0.944	500	0
500	0.75	0.10	Oracle-GMM	-0.25	0.050	0.950	500	0
500	0.75	0.10	Oracle-GMM	0.25	0.072	0.928	500	0
500	0.75	0.10	Oracle-GMM	0.50	0.076	0.924	500	0
500	0.75	0.10	Oracle-GMM	1.00	0.104	0.896	500	0
500	0.75	0.10	Full-GMM	-1.00	0.050	0.950	500	0
500	0.75	0.10	Full-GMM	-0.50	0.052	0.948	500	0
500	0.75	0.10	Full-GMM	-0.25	0.042	0.958	500	0
500	0.75	0.10	Full-GMM	0.25	0.042	0.958	500	0
500	0.75	0.10	Full-GMM	0.50	0.068	0.932	500	0
500	0.75	0.10	Full-GMM	1.00	0.068	0.932	500	0
500	0.75	0.10	DML-IVQR-BC	-1.00	0.102	0.898	500	0
500	0.75	0.10	DML-IVQR-BC	-0.50	0.076	0.924	500	0
500	0.75	0.10	DML-IVQR-BC	-0.25	0.056	0.944	500	0
500	0.75	0.10	DML-IVQR-BC	0.25	0.050	0.950	500	0
500	0.75	0.10	DML-IVQR-BC	0.50	0.044	0.956	500	0
500	0.75	0.10	DML-IVQR-BC	1.00	0.058	0.942	500	0
500	0.90	1.00	Oracle-GMM	-1.00	0.658	0.342	500	0
500	0.90	1.00	Oracle-GMM	-0.50	0.278	0.722	500	0
500	0.90	1.00	Oracle-GMM	-0.25	0.096	0.904	500	0
500	0.90	1.00	Oracle-GMM	0.25	0.584	0.416	500	0
500	0.90	1.00	Oracle-GMM	0.50	0.954	0.046	500	0
500	0.90	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
500	0.90	1.00	Full-GMM	-1.00	0.098	0.902	500	0
500	0.90	1.00	Full-GMM	-0.50	0.068	0.932	500	0
500	0.90	1.00	Full-GMM	-0.25	0.186	0.814	500	0
500	0.90	1.00	Full-GMM	0.25	0.724	0.276	500	0
500	0.90	1.00	Full-GMM	0.50	0.834	0.166	500	0
500	0.90	1.00	Full-GMM	1.00	0.960	0.040	500	0
500	0.90	1.00	DML-IVQR-BC	-1.00	0.832	0.168	500	0
500	0.90	1.00	DML-IVQR-BC	-0.50	0.262	0.738	500	0
500	0.90	1.00	DML-IVQR-BC	-0.25	0.072	0.928	500	0
500	0.90	1.00	DML-IVQR-BC	0.25	0.882	0.118	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.90	1.00	DML-IVQR-BC	0.50	0.998	0.002	500	0
500	0.90	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
500	0.90	0.50	Oracle-GMM	-1.00	0.166	0.834	500	0
500	0.90	0.50	Oracle-GMM	-0.50	0.088	0.912	500	0
500	0.90	0.50	Oracle-GMM	-0.25	0.052	0.948	500	0
500	0.90	0.50	Oracle-GMM	0.25	0.212	0.788	500	0
500	0.90	0.50	Oracle-GMM	0.50	0.446	0.554	500	0
500	0.90	0.50	Oracle-GMM	1.00	0.794	0.206	500	0
500	0.90	0.50	Full-GMM	-1.00	0.038	0.962	500	0
500	0.90	0.50	Full-GMM	-0.50	0.048	0.952	500	0
500	0.90	0.50	Full-GMM	-0.25	0.072	0.928	500	0
500	0.90	0.50	Full-GMM	0.25	0.210	0.790	500	0
500	0.90	0.50	Full-GMM	0.50	0.324	0.676	500	0
500	0.90	0.50	Full-GMM	1.00	0.524	0.476	500	0
500	0.90	0.50	DML-IVQR-BC	-1.00	0.300	0.700	500	0
500	0.90	0.50	DML-IVQR-BC	-0.50	0.102	0.898	500	0
500	0.90	0.50	DML-IVQR-BC	-0.25	0.054	0.946	500	0
500	0.90	0.50	DML-IVQR-BC	0.25	0.314	0.686	500	0
500	0.90	0.50	DML-IVQR-BC	0.50	0.530	0.470	500	0
500	0.90	0.50	DML-IVQR-BC	1.00	0.766	0.234	500	0
500	0.90	0.25	Oracle-GMM	-1.00	0.078	0.922	500	0
500	0.90	0.25	Oracle-GMM	-0.50	0.044	0.956	500	0
500	0.90	0.25	Oracle-GMM	-0.25	0.042	0.958	500	0
500	0.90	0.25	Oracle-GMM	0.25	0.102	0.898	500	0
500	0.90	0.25	Oracle-GMM	0.50	0.162	0.838	500	0
500	0.90	0.25	Oracle-GMM	1.00	0.260	0.740	500	0
500	0.90	0.25	Full-GMM	-1.00	0.058	0.942	500	0
500	0.90	0.25	Full-GMM	-0.50	0.038	0.962	500	0
500	0.90	0.25	Full-GMM	-0.25	0.050	0.950	500	0
500	0.90	0.25	Full-GMM	0.25	0.086	0.914	500	0
500	0.90	0.25	Full-GMM	0.50	0.116	0.884	500	0
500	0.90	0.25	Full-GMM	1.00	0.172	0.828	500	0
500	0.90	0.25	DML-IVQR-BC	-1.00	0.110	0.890	500	0
500	0.90	0.25	DML-IVQR-BC	-0.50	0.086	0.914	500	0
500	0.90	0.25	DML-IVQR-BC	-0.25	0.064	0.936	500	0
500	0.90	0.25	DML-IVQR-BC	0.25	0.100	0.900	500	0
500	0.90	0.25	DML-IVQR-BC	0.50	0.154	0.846	500	0
500	0.90	0.25	DML-IVQR-BC	1.00	0.174	0.826	500	0
500	0.90	0.10	Oracle-GMM	-1.00	0.062	0.938	500	0
500	0.90	0.10	Oracle-GMM	-0.50	0.046	0.954	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
500	0.90	0.10	Oracle-GMM	-0.25	0.032	0.968	500	0
500	0.90	0.10	Oracle-GMM	0.25	0.058	0.942	500	0
500	0.90	0.10	Oracle-GMM	0.50	0.068	0.932	500	0
500	0.90	0.10	Oracle-GMM	1.00	0.096	0.904	500	0
500	0.90	0.10	Full-GMM	-1.00	0.026	0.974	500	0
500	0.90	0.10	Full-GMM	-0.50	0.032	0.968	500	0
500	0.90	0.10	Full-GMM	-0.25	0.042	0.958	500	0
500	0.90	0.10	Full-GMM	0.25	0.058	0.942	500	0
500	0.90	0.10	Full-GMM	0.50	0.052	0.948	500	0
500	0.90	0.10	Full-GMM	1.00	0.058	0.942	500	0
500	0.90	0.10	DML-IVQR-BC	-1.00	0.050	0.950	500	0
500	0.90	0.10	DML-IVQR-BC	-0.50	0.064	0.936	500	0
500	0.90	0.10	DML-IVQR-BC	-0.25	0.056	0.944	500	0
500	0.90	0.10	DML-IVQR-BC	0.25	0.060	0.940	500	0
500	0.90	0.10	DML-IVQR-BC	0.50	0.068	0.932	500	0
500	0.90	0.10	DML-IVQR-BC	1.00	0.040	0.960	500	0
1000	0.10	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.10	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.10	1.00	Oracle-GMM	-0.25	0.938	0.062	500	0
1000	0.10	1.00	Oracle-GMM	0.25	0.462	0.538	500	0
1000	0.10	1.00	Oracle-GMM	0.50	0.930	0.070	500	0
1000	0.10	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.10	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.10	1.00	Full-GMM	-0.50	0.994	0.006	500	0
1000	0.10	1.00	Full-GMM	-0.25	0.882	0.118	500	0
1000	0.10	1.00	Full-GMM	0.25	0.100	0.900	500	0
1000	0.10	1.00	Full-GMM	0.50	0.126	0.874	500	0
1000	0.10	1.00	Full-GMM	1.00	0.822	0.178	500	0
1000	0.10	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.10	1.00	DML-IVQR-BC	-0.50	0.988	0.012	500	0
1000	0.10	1.00	DML-IVQR-BC	-0.25	0.482	0.518	500	0
1000	0.10	1.00	DML-IVQR-BC	0.25	0.296	0.704	500	0
1000	0.10	1.00	DML-IVQR-BC	0.50	0.696	0.304	500	0
1000	0.10	1.00	DML-IVQR-BC	1.00	0.966	0.034	500	0
1000	0.10	0.50	Oracle-GMM	-1.00	0.988	0.012	500	0
1000	0.10	0.50	Oracle-GMM	-0.50	0.852	0.148	500	0
1000	0.10	0.50	Oracle-GMM	-0.25	0.438	0.562	500	0
1000	0.10	0.50	Oracle-GMM	0.25	0.156	0.844	500	0
1000	0.10	0.50	Oracle-GMM	0.50	0.364	0.636	500	0
1000	0.10	0.50	Oracle-GMM	1.00	0.662	0.338	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.10	0.50	Full-GMM	-1.00	0.914	0.086	500	0
1000	0.10	0.50	Full-GMM	-0.50	0.682	0.318	500	0
1000	0.10	0.50	Full-GMM	-0.25	0.440	0.560	500	0
1000	0.10	0.50	Full-GMM	0.25	0.052	0.948	500	0
1000	0.10	0.50	Full-GMM	0.50	0.046	0.954	500	0
1000	0.10	0.50	Full-GMM	1.00	0.274	0.726	500	0
1000	0.10	0.50	DML-IVQR-BC	-1.00	0.976	0.024	500	0
1000	0.10	0.50	DML-IVQR-BC	-0.50	0.678	0.322	500	0
1000	0.10	0.50	DML-IVQR-BC	-0.25	0.202	0.798	500	0
1000	0.10	0.50	DML-IVQR-BC	0.25	0.066	0.934	500	0
1000	0.10	0.50	DML-IVQR-BC	0.50	0.148	0.852	500	0
1000	0.10	0.50	DML-IVQR-BC	1.00	0.336	0.664	500	0
1000	0.10	0.25	Oracle-GMM	-1.00	0.526	0.474	500	0
1000	0.10	0.25	Oracle-GMM	-0.50	0.342	0.658	500	0
1000	0.10	0.25	Oracle-GMM	-0.25	0.154	0.846	500	0
1000	0.10	0.25	Oracle-GMM	0.25	0.078	0.922	500	0
1000	0.10	0.25	Oracle-GMM	0.50	0.110	0.890	500	0
1000	0.10	0.25	Oracle-GMM	1.00	0.194	0.806	500	0
1000	0.10	0.25	Full-GMM	-1.00	0.392	0.608	500	0
1000	0.10	0.25	Full-GMM	-0.50	0.230	0.770	500	0
1000	0.10	0.25	Full-GMM	-0.25	0.144	0.856	500	0
1000	0.10	0.25	Full-GMM	0.25	0.044	0.956	500	0
1000	0.10	0.25	Full-GMM	0.50	0.050	0.950	500	0
1000	0.10	0.25	Full-GMM	1.00	0.096	0.904	500	0
1000	0.10	0.25	DML-IVQR-BC	-1.00	0.608	0.392	500	0
1000	0.10	0.25	DML-IVQR-BC	-0.50	0.306	0.694	500	0
1000	0.10	0.25	DML-IVQR-BC	-0.25	0.098	0.902	500	0
1000	0.10	0.25	DML-IVQR-BC	0.25	0.028	0.972	500	0
1000	0.10	0.25	DML-IVQR-BC	0.50	0.036	0.964	500	0
1000	0.10	0.25	DML-IVQR-BC	1.00	0.058	0.942	500	0
1000	0.10	0.10	Oracle-GMM	-1.00	0.132	0.868	500	0
1000	0.10	0.10	Oracle-GMM	-0.50	0.104	0.896	500	0
1000	0.10	0.10	Oracle-GMM	-0.25	0.078	0.922	500	0
1000	0.10	0.10	Oracle-GMM	0.25	0.056	0.944	500	0
1000	0.10	0.10	Oracle-GMM	0.50	0.058	0.942	500	0
1000	0.10	0.10	Oracle-GMM	1.00	0.052	0.948	500	0
1000	0.10	0.10	Full-GMM	-1.00	0.096	0.904	500	0
1000	0.10	0.10	Full-GMM	-0.50	0.080	0.920	500	0
1000	0.10	0.10	Full-GMM	-0.25	0.054	0.946	500	0
1000	0.10	0.10	Full-GMM	0.25	0.056	0.944	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.10	0.10	Full-GMM	0.50	0.050	0.950	500	0
1000	0.10	0.10	Full-GMM	1.00	0.046	0.954	500	0
1000	0.10	0.10	DML-IVQR-BC	-1.00	0.242	0.758	500	0
1000	0.10	0.10	DML-IVQR-BC	-0.50	0.118	0.882	500	0
1000	0.10	0.10	DML-IVQR-BC	-0.25	0.076	0.924	500	0
1000	0.10	0.10	DML-IVQR-BC	0.25	0.038	0.962	500	0
1000	0.10	0.10	DML-IVQR-BC	0.50	0.022	0.978	500	0
1000	0.10	0.10	DML-IVQR-BC	1.00	0.024	0.976	500	0
1000	0.25	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	-0.25	0.980	0.020	500	0
1000	0.25	1.00	Oracle-GMM	0.25	0.898	0.102	500	0
1000	0.25	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.25	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.25	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.25	1.00	Full-GMM	-0.50	0.998	0.002	500	0
1000	0.25	1.00	Full-GMM	-0.25	0.904	0.096	500	0
1000	0.25	1.00	Full-GMM	0.25	0.282	0.718	500	0
1000	0.25	1.00	Full-GMM	0.50	0.894	0.106	500	0
1000	0.25	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-0.50	1.000	0.000	500	0
1000	0.25	1.00	DML-IVQR-BC	-0.25	0.746	0.254	500	0
1000	0.25	1.00	DML-IVQR-BC	0.25	0.838	0.162	500	0
1000	0.25	1.00	DML-IVQR-BC	0.50	0.996	0.004	500	0
1000	0.25	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.25	0.50	Oracle-GMM	-1.00	0.998	0.002	500	0
1000	0.25	0.50	Oracle-GMM	-0.50	0.902	0.098	500	0
1000	0.25	0.50	Oracle-GMM	-0.25	0.542	0.458	500	0
1000	0.25	0.50	Oracle-GMM	0.25	0.318	0.682	500	0
1000	0.25	0.50	Oracle-GMM	0.50	0.758	0.242	500	0
1000	0.25	0.50	Oracle-GMM	1.00	0.968	0.032	500	0
1000	0.25	0.50	Full-GMM	-1.00	0.986	0.014	500	0
1000	0.25	0.50	Full-GMM	-0.50	0.782	0.218	500	0
1000	0.25	0.50	Full-GMM	-0.25	0.418	0.582	500	0
1000	0.25	0.50	Full-GMM	0.25	0.082	0.918	500	0
1000	0.25	0.50	Full-GMM	0.50	0.350	0.650	500	0
1000	0.25	0.50	Full-GMM	1.00	0.824	0.176	500	0
1000	0.25	0.50	DML-IVQR-BC	-1.00	0.998	0.002	500	0
1000	0.25	0.50	DML-IVQR-BC	-0.50	0.832	0.168	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.25	0.50	DML-IVQR-BC	-0.25	0.310	0.690	500	0
1000	0.25	0.50	DML-IVQR-BC	0.25	0.262	0.738	500	0
1000	0.25	0.50	DML-IVQR-BC	0.50	0.556	0.444	500	0
1000	0.25	0.50	DML-IVQR-BC	1.00	0.848	0.152	500	0
1000	0.25	0.25	Oracle-GMM	-1.00	0.610	0.390	500	0
1000	0.25	0.25	Oracle-GMM	-0.50	0.386	0.614	500	0
1000	0.25	0.25	Oracle-GMM	-0.25	0.188	0.812	500	0
1000	0.25	0.25	Oracle-GMM	0.25	0.124	0.876	500	0
1000	0.25	0.25	Oracle-GMM	0.50	0.224	0.776	500	0
1000	0.25	0.25	Oracle-GMM	1.00	0.416	0.584	500	0
1000	0.25	0.25	Full-GMM	-1.00	0.512	0.488	500	0
1000	0.25	0.25	Full-GMM	-0.50	0.322	0.678	500	0
1000	0.25	0.25	Full-GMM	-0.25	0.128	0.872	500	0
1000	0.25	0.25	Full-GMM	0.25	0.052	0.948	500	0
1000	0.25	0.25	Full-GMM	0.50	0.112	0.888	500	0
1000	0.25	0.25	Full-GMM	1.00	0.308	0.692	500	0
1000	0.25	0.25	DML-IVQR-BC	-1.00	0.688	0.312	500	0
1000	0.25	0.25	DML-IVQR-BC	-0.50	0.362	0.638	500	0
1000	0.25	0.25	DML-IVQR-BC	-0.25	0.156	0.844	500	0
1000	0.25	0.25	DML-IVQR-BC	0.25	0.084	0.916	500	0
1000	0.25	0.25	DML-IVQR-BC	0.50	0.136	0.864	500	0
1000	0.25	0.25	DML-IVQR-BC	1.00	0.260	0.740	500	0
1000	0.25	0.10	Oracle-GMM	-1.00	0.170	0.830	500	0
1000	0.25	0.10	Oracle-GMM	-0.50	0.106	0.894	500	0
1000	0.25	0.10	Oracle-GMM	-0.25	0.086	0.914	500	0
1000	0.25	0.10	Oracle-GMM	0.25	0.064	0.936	500	0
1000	0.25	0.10	Oracle-GMM	0.50	0.086	0.914	500	0
1000	0.25	0.10	Oracle-GMM	1.00	0.094	0.906	500	0
1000	0.25	0.10	Full-GMM	-1.00	0.126	0.874	500	0
1000	0.25	0.10	Full-GMM	-0.50	0.088	0.912	500	0
1000	0.25	0.10	Full-GMM	-0.25	0.056	0.944	500	0
1000	0.25	0.10	Full-GMM	0.25	0.052	0.948	500	0
1000	0.25	0.10	Full-GMM	0.50	0.066	0.934	500	0
1000	0.25	0.10	Full-GMM	1.00	0.096	0.904	500	0
1000	0.25	0.10	DML-IVQR-BC	-1.00	0.220	0.780	500	0
1000	0.25	0.10	DML-IVQR-BC	-0.50	0.120	0.880	500	0
1000	0.25	0.10	DML-IVQR-BC	-0.25	0.090	0.910	500	0
1000	0.25	0.10	DML-IVQR-BC	0.25	0.044	0.956	500	0
1000	0.25	0.10	DML-IVQR-BC	0.50	0.060	0.940	500	0
1000	0.25	0.10	DML-IVQR-BC	1.00	0.056	0.944	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.50	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	-0.50	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	-0.25	0.948	0.052	500	0
1000	0.50	1.00	Oracle-GMM	0.25	0.980	0.020	500	0
1000	0.50	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.50	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.50	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.50	1.00	Full-GMM	-0.50	0.998	0.002	500	0
1000	0.50	1.00	Full-GMM	-0.25	0.692	0.308	500	0
1000	0.50	1.00	Full-GMM	0.25	0.754	0.246	500	0
1000	0.50	1.00	Full-GMM	0.50	0.998	0.002	500	0
1000	0.50	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-0.50	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	-0.25	0.774	0.226	500	0
1000	0.50	1.00	DML-IVQR-BC	0.25	0.988	0.012	500	0
1000	0.50	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.50	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.50	0.50	Oracle-GMM	-1.00	0.986	0.014	500	0
1000	0.50	0.50	Oracle-GMM	-0.50	0.850	0.150	500	0
1000	0.50	0.50	Oracle-GMM	-0.25	0.426	0.574	500	0
1000	0.50	0.50	Oracle-GMM	0.25	0.540	0.460	500	0
1000	0.50	0.50	Oracle-GMM	0.50	0.940	0.060	500	0
1000	0.50	0.50	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.50	0.50	Full-GMM	-1.00	0.952	0.048	500	0
1000	0.50	0.50	Full-GMM	-0.50	0.634	0.366	500	0
1000	0.50	0.50	Full-GMM	-0.25	0.242	0.758	500	0
1000	0.50	0.50	Full-GMM	0.25	0.298	0.702	500	0
1000	0.50	0.50	Full-GMM	0.50	0.726	0.274	500	0
1000	0.50	0.50	Full-GMM	1.00	0.988	0.012	500	0
1000	0.50	0.50	DML-IVQR-BC	-1.00	0.994	0.006	500	0
1000	0.50	0.50	DML-IVQR-BC	-0.50	0.802	0.198	500	0
1000	0.50	0.50	DML-IVQR-BC	-0.25	0.284	0.716	500	0
1000	0.50	0.50	DML-IVQR-BC	0.25	0.494	0.506	500	0
1000	0.50	0.50	DML-IVQR-BC	0.50	0.852	0.148	500	0
1000	0.50	0.50	DML-IVQR-BC	1.00	0.994	0.006	500	0
1000	0.50	0.25	Oracle-GMM	-1.00	0.540	0.460	500	0
1000	0.50	0.25	Oracle-GMM	-0.50	0.316	0.684	500	0
1000	0.50	0.25	Oracle-GMM	-0.25	0.124	0.876	500	0
1000	0.50	0.25	Oracle-GMM	0.25	0.190	0.810	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.50	0.25	Oracle-GMM	0.50	0.378	0.622	500	0
1000	0.50	0.25	Oracle-GMM	1.00	0.626	0.374	500	0
1000	0.50	0.25	Full-GMM	-1.00	0.430	0.570	500	0
1000	0.50	0.25	Full-GMM	-0.50	0.212	0.788	500	0
1000	0.50	0.25	Full-GMM	-0.25	0.092	0.908	500	0
1000	0.50	0.25	Full-GMM	0.25	0.102	0.898	500	0
1000	0.50	0.25	Full-GMM	0.50	0.238	0.762	500	0
1000	0.50	0.25	Full-GMM	1.00	0.524	0.476	500	0
1000	0.50	0.25	DML-IVQR-BC	-1.00	0.632	0.368	500	0
1000	0.50	0.25	DML-IVQR-BC	-0.50	0.312	0.688	500	0
1000	0.50	0.25	DML-IVQR-BC	-0.25	0.108	0.892	500	0
1000	0.50	0.25	DML-IVQR-BC	0.25	0.152	0.848	500	0
1000	0.50	0.25	DML-IVQR-BC	0.50	0.286	0.714	500	0
1000	0.50	0.25	DML-IVQR-BC	1.00	0.486	0.514	500	0
1000	0.50	0.10	Oracle-GMM	-1.00	0.120	0.880	500	0
1000	0.50	0.10	Oracle-GMM	-0.50	0.082	0.918	500	0
1000	0.50	0.10	Oracle-GMM	-0.25	0.064	0.936	500	0
1000	0.50	0.10	Oracle-GMM	0.25	0.084	0.916	500	0
1000	0.50	0.10	Oracle-GMM	0.50	0.092	0.908	500	0
1000	0.50	0.10	Oracle-GMM	1.00	0.138	0.862	500	0
1000	0.50	0.10	Full-GMM	-1.00	0.112	0.888	500	0
1000	0.50	0.10	Full-GMM	-0.50	0.076	0.924	500	0
1000	0.50	0.10	Full-GMM	-0.25	0.054	0.946	500	0
1000	0.50	0.10	Full-GMM	0.25	0.066	0.934	500	0
1000	0.50	0.10	Full-GMM	0.50	0.098	0.902	500	0
1000	0.50	0.10	Full-GMM	1.00	0.160	0.840	500	0
1000	0.50	0.10	DML-IVQR-BC	-1.00	0.154	0.846	500	0
1000	0.50	0.10	DML-IVQR-BC	-0.50	0.104	0.896	500	0
1000	0.50	0.10	DML-IVQR-BC	-0.25	0.060	0.940	500	0
1000	0.50	0.10	DML-IVQR-BC	0.25	0.070	0.930	500	0
1000	0.50	0.10	DML-IVQR-BC	0.50	0.094	0.906	500	0
1000	0.50	0.10	DML-IVQR-BC	1.00	0.080	0.920	500	0
1000	0.75	1.00	Oracle-GMM	-1.00	1.000	0.000	500	0
1000	0.75	1.00	Oracle-GMM	-0.50	0.994	0.006	500	0
1000	0.75	1.00	Oracle-GMM	-0.25	0.746	0.254	500	0
1000	0.75	1.00	Oracle-GMM	0.25	0.976	0.024	500	0
1000	0.75	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.75	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	-1.00	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	-0.50	0.760	0.240	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.75	1.00	Full-GMM	-0.25	0.162	0.838	500	0
1000	0.75	1.00	Full-GMM	0.25	0.906	0.094	500	0
1000	0.75	1.00	Full-GMM	0.50	1.000	0.000	500	0
1000	0.75	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	-1.00	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	-0.50	0.984	0.016	500	0
1000	0.75	1.00	DML-IVQR-BC	-0.25	0.510	0.490	500	0
1000	0.75	1.00	DML-IVQR-BC	0.25	0.996	0.004	500	0
1000	0.75	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.75	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.75	0.50	Oracle-GMM	-1.00	0.828	0.172	500	0
1000	0.75	0.50	Oracle-GMM	-0.50	0.526	0.474	500	0
1000	0.75	0.50	Oracle-GMM	-0.25	0.260	0.740	500	0
1000	0.75	0.50	Oracle-GMM	0.25	0.554	0.446	500	0
1000	0.75	0.50	Oracle-GMM	0.50	0.950	0.050	500	0
1000	0.75	0.50	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.75	0.50	Full-GMM	-1.00	0.654	0.346	500	0
1000	0.75	0.50	Full-GMM	-0.50	0.230	0.770	500	0
1000	0.75	0.50	Full-GMM	-0.25	0.078	0.922	500	0
1000	0.75	0.50	Full-GMM	0.25	0.414	0.586	500	0
1000	0.75	0.50	Full-GMM	0.50	0.786	0.214	500	0
1000	0.75	0.50	Full-GMM	1.00	0.996	0.004	500	0
1000	0.75	0.50	DML-IVQR-BC	-1.00	0.906	0.094	500	0
1000	0.75	0.50	DML-IVQR-BC	-0.50	0.546	0.454	500	0
1000	0.75	0.50	DML-IVQR-BC	-0.25	0.184	0.816	500	0
1000	0.75	0.50	DML-IVQR-BC	0.25	0.640	0.360	500	0
1000	0.75	0.50	DML-IVQR-BC	0.50	0.958	0.042	500	0
1000	0.75	0.50	DML-IVQR-BC	1.00	0.998	0.002	500	0
1000	0.75	0.25	Oracle-GMM	-1.00	0.286	0.714	500	0
1000	0.75	0.25	Oracle-GMM	-0.50	0.154	0.846	500	0
1000	0.75	0.25	Oracle-GMM	-0.25	0.084	0.916	500	0
1000	0.75	0.25	Oracle-GMM	0.25	0.220	0.780	500	0
1000	0.75	0.25	Oracle-GMM	0.50	0.418	0.582	500	0
1000	0.75	0.25	Oracle-GMM	1.00	0.712	0.288	500	0
1000	0.75	0.25	Full-GMM	-1.00	0.192	0.808	500	0
1000	0.75	0.25	Full-GMM	-0.50	0.084	0.916	500	0
1000	0.75	0.25	Full-GMM	-0.25	0.040	0.960	500	0
1000	0.75	0.25	Full-GMM	0.25	0.152	0.848	500	0
1000	0.75	0.25	Full-GMM	0.50	0.292	0.708	500	0
1000	0.75	0.25	Full-GMM	1.00	0.548	0.452	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	**Estimator**	Δ	**Power**	**False acc.**	**Successful**	**Failed**
1000	0.75	0.25	DML-IVQR-BC	-1.00	0.398	0.602	500	0
1000	0.75	0.25	DML-IVQR-BC	-0.50	0.194	0.806	500	0
1000	0.75	0.25	DML-IVQR-BC	-0.25	0.100	0.900	500	0
1000	0.75	0.25	DML-IVQR-BC	0.25	0.212	0.788	500	0
1000	0.75	0.25	DML-IVQR-BC	0.50	0.384	0.616	500	0
1000	0.75	0.25	DML-IVQR-BC	1.00	0.610	0.390	500	0
1000	0.75	0.10	Oracle-GMM	-1.00	0.088	0.912	500	0
1000	0.75	0.10	Oracle-GMM	-0.50	0.054	0.946	500	0
1000	0.75	0.10	Oracle-GMM	-0.25	0.046	0.954	500	0
1000	0.75	0.10	Oracle-GMM	0.25	0.070	0.930	500	0
1000	0.75	0.10	Oracle-GMM	0.50	0.142	0.858	500	0
1000	0.75	0.10	Oracle-GMM	1.00	0.166	0.834	500	0
1000	0.75	0.10	Full-GMM	-1.00	0.070	0.930	500	0
1000	0.75	0.10	Full-GMM	-0.50	0.064	0.936	500	0
1000	0.75	0.10	Full-GMM	-0.25	0.070	0.930	500	0
1000	0.75	0.10	Full-GMM	0.25	0.074	0.926	500	0
1000	0.75	0.10	Full-GMM	0.50	0.096	0.904	500	0
1000	0.75	0.10	Full-GMM	1.00	0.122	0.878	500	0
1000	0.75	0.10	DML-IVQR-BC	-1.00	0.102	0.898	500	0
1000	0.75	0.10	DML-IVQR-BC	-0.50	0.090	0.910	500	0
1000	0.75	0.10	DML-IVQR-BC	-0.25	0.080	0.920	500	0
1000	0.75	0.10	DML-IVQR-BC	0.25	0.068	0.932	500	0
1000	0.75	0.10	DML-IVQR-BC	0.50	0.100	0.900	500	0
1000	0.75	0.10	DML-IVQR-BC	1.00	0.118	0.882	500	0
1000	0.90	1.00	Oracle-GMM	-1.00	0.972	0.028	500	0
1000	0.90	1.00	Oracle-GMM	-0.50	0.748	0.252	500	0
1000	0.90	1.00	Oracle-GMM	-0.25	0.322	0.678	500	0
1000	0.90	1.00	Oracle-GMM	0.25	0.928	0.072	500	0
1000	0.90	1.00	Oracle-GMM	0.50	1.000	0.000	500	0
1000	0.90	1.00	Oracle-GMM	1.00	1.000	0.000	500	0
1000	0.90	1.00	Full-GMM	-1.00	0.544	0.456	500	0
1000	0.90	1.00	Full-GMM	-0.50	0.112	0.888	500	0
1000	0.90	1.00	Full-GMM	-0.25	0.114	0.886	500	0
1000	0.90	1.00	Full-GMM	0.25	0.840	0.160	500	0
1000	0.90	1.00	Full-GMM	0.50	0.982	0.018	500	0
1000	0.90	1.00	Full-GMM	1.00	1.000	0.000	500	0
1000	0.90	1.00	DML-IVQR-BC	-1.00	0.990	0.010	500	0
1000	0.90	1.00	DML-IVQR-BC	-0.50	0.660	0.340	500	0
1000	0.90	1.00	DML-IVQR-BC	-0.25	0.128	0.872	500	0
1000	0.90	1.00	DML-IVQR-BC	0.25	0.996	0.004	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.90	1.00	DML-IVQR-BC	0.50	1.000	0.000	500	0
1000	0.90	1.00	DML-IVQR-BC	1.00	1.000	0.000	500	0
1000	0.90	0.50	Oracle-GMM	-1.00	0.426	0.574	500	0
1000	0.90	0.50	Oracle-GMM	-0.50	0.228	0.772	500	0
1000	0.90	0.50	Oracle-GMM	-0.25	0.070	0.930	500	0
1000	0.90	0.50	Oracle-GMM	0.25	0.440	0.560	500	0
1000	0.90	0.50	Oracle-GMM	0.50	0.844	0.156	500	0
1000	0.90	0.50	Oracle-GMM	1.00	0.998	0.002	500	0
1000	0.90	0.50	Full-GMM	-1.00	0.172	0.828	500	0
1000	0.90	0.50	Full-GMM	-0.50	0.056	0.944	500	0
1000	0.90	0.50	Full-GMM	-0.25	0.074	0.926	500	0
1000	0.90	0.50	Full-GMM	0.25	0.400	0.600	500	0
1000	0.90	0.50	Full-GMM	0.50	0.672	0.328	500	0
1000	0.90	0.50	Full-GMM	1.00	0.936	0.064	500	0
1000	0.90	0.50	DML-IVQR-BC	-1.00	0.578	0.422	500	0
1000	0.90	0.50	DML-IVQR-BC	-0.50	0.224	0.776	500	0
1000	0.90	0.50	DML-IVQR-BC	-0.25	0.070	0.930	500	0
1000	0.90	0.50	DML-IVQR-BC	0.25	0.616	0.384	500	0
1000	0.90	0.50	DML-IVQR-BC	0.50	0.900	0.100	500	0
1000	0.90	0.50	DML-IVQR-BC	1.00	0.996	0.004	500	0
1000	0.90	0.25	Oracle-GMM	-1.00	0.144	0.856	500	0
1000	0.90	0.25	Oracle-GMM	-0.50	0.086	0.914	500	0
1000	0.90	0.25	Oracle-GMM	-0.25	0.052	0.948	500	0
1000	0.90	0.25	Oracle-GMM	0.25	0.166	0.834	500	0
1000	0.90	0.25	Oracle-GMM	0.50	0.338	0.662	500	0
1000	0.90	0.25	Oracle-GMM	1.00	0.558	0.442	500	0
1000	0.90	0.25	Full-GMM	-1.00	0.066	0.934	500	0
1000	0.90	0.25	Full-GMM	-0.50	0.050	0.950	500	0
1000	0.90	0.25	Full-GMM	-0.25	0.050	0.950	500	0
1000	0.90	0.25	Full-GMM	0.25	0.126	0.874	500	0
1000	0.90	0.25	Full-GMM	0.50	0.258	0.742	500	0
1000	0.90	0.25	Full-GMM	1.00	0.416	0.584	500	0
1000	0.90	0.25	DML-IVQR-BC	-1.00	0.208	0.792	500	0
1000	0.90	0.25	DML-IVQR-BC	-0.50	0.126	0.874	500	0
1000	0.90	0.25	DML-IVQR-BC	-0.25	0.062	0.938	500	0
1000	0.90	0.25	DML-IVQR-BC	0.25	0.194	0.806	500	0
1000	0.90	0.25	DML-IVQR-BC	0.50	0.306	0.694	500	0
1000	0.90	0.25	DML-IVQR-BC	1.00	0.490	0.510	500	0
1000	0.90	0.10	Oracle-GMM	-1.00	0.080	0.920	500	0
1000	0.90	0.10	Oracle-GMM	-0.50	0.062	0.938	500	0

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TableB.1. Full power results (Continued)

n	τ	κ	Estimator	Δ	Power	False acc.	Successful	Failed
1000	0.90	0.10	Oracle-GMM	-0.25	0.038	0.962	500	0
1000	0.90	0.10	Oracle-GMM	0.25	0.092	0.908	500	0
1000	0.90	0.10	Oracle-GMM	0.50	0.096	0.904	500	0
1000	0.90	0.10	Oracle-GMM	1.00	0.144	0.856	500	0
1000	0.90	0.10	Full-GMM	-1.00	0.052	0.948	500	0
1000	0.90	0.10	Full-GMM	-0.50	0.034	0.966	500	0
1000	0.90	0.10	Full-GMM	-0.25	0.042	0.958	500	0
1000	0.90	0.10	Full-GMM	0.25	0.066	0.934	500	0
1000	0.90	0.10	Full-GMM	0.50	0.060	0.940	500	0
1000	0.90	0.10	Full-GMM	1.00	0.088	0.912	500	0
1000	0.90	0.10	DML-IVQR-BC	-1.00	0.096	0.904	500	0
1000	0.90	0.10	DML-IVQR-BC	-0.50	0.064	0.936	500	0
1000	0.90	0.10	DML-IVQR-BC	-0.25	0.054	0.946	500	0
1000	0.90	0.10	DML-IVQR-BC	0.25	0.082	0.918	500	0
1000	0.90	0.10	DML-IVQR-BC	0.50	0.058	0.942	500	0
1000	0.90	0.10	DML-IVQR-BC	1.00	0.100	0.900	500	0

Note: Rejection probabilities at false values $a = \alpha_0(\tau) + \Delta$. These probabilities are power, not empirical size.

Appendix C Boundary and Grid-Domain Diagnostic

Table C.1. Grid-domain diagnostic: accepted-set measures

n	τ	κ	Estimator	Med. A_0	Med. A_1	Med. A_2	Mean A_0	Mean A_1	Mean A_2
1000	0.10	1.00	Oracle-GMM	0.400	0.400	0.400	0.386	0.386	0.386
1000	0.10	1.00	Full-GMM	0.800	0.800	0.800	0.757	0.757	0.757
1000	0.10	1.00	DML-IVQR-BC	0.600	0.600	0.600	0.705	0.705	0.705
1000	0.10	0.25	Oracle-GMM	3.400	5.325	7.250	3.019	4.994	6.838
1000	0.10	0.25	Full-GMM	3.375	5.950	8.200	3.304	5.774	8.037
1000	0.10	0.25	DML-IVQR-BC	3.750	5.950	7.950	3.607	6.040	8.245
1000	0.10	0.10	Oracle-GMM	4.000	7.900	11.900	3.745	7.381	11.002
1000	0.10	0.10	Full-GMM	3.900	7.800	11.600	3.785	7.394	10.975
1000	0.10	0.10	DML-IVQR-BC	4.000	7.575	10.775	3.845	7.219	10.339
1000	0.25	1.00	Oracle-GMM	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.25	1.00	Full-GMM	0.400	0.400	0.400	0.429	0.429	0.429
1000	0.25	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.346	0.346	0.346

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TableC.1. Grid-domain diagnostic: accepted-set measures (Continued)

n	τ	κ	**Estimator**	**Med.** A_0	**Med.** A_1	**Med.** A_2	**Mean** A_0	**Mean** A_1	**Mean** A_2
1000	0.25	0.25	Oracle-GMM	2.175	2.800	2.850	2.224	3.158	3.964
1000	0.25	0.25	Full-GMM	2.450	3.500	4.075	2.517	3.768	4.745
1000	0.25	0.25	DML-IVQR-BC	2.750	4.650	6.225	2.630	4.130	5.712
1000	0.25	0.10	Oracle-GMM	3.900	7.700	11.500	3.492	6.699	9.872
1000	0.25	0.10	Full-GMM	3.825	7.600	11.100	3.595	6.891	10.083
1000	0.25	0.10	DML-IVQR-BC	4.000	7.700	11.125	3.651	6.926	9.984
1000	0.50	1.00	Oracle-GMM	0.200	0.200	0.200	0.235	0.235	0.235
1000	0.50	1.00	Full-GMM	0.400	0.400	0.400	0.373	0.373	0.373
1000	0.50	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.259	0.259	0.259
1000	0.50	0.25	Oracle-GMM	1.875	2.200	2.200	1.960	2.682	3.229
1000	0.50	0.25	Full-GMM	2.250	2.400	2.550	2.368	3.171	3.802
1000	0.50	0.25	DML-IVQR-BC	1.950	2.500	3.025	2.124	3.020	3.942
1000	0.50	0.10	Oracle-GMM	3.825	7.600	11.300	3.521	6.640	9.687
1000	0.50	0.10	Full-GMM	3.900	7.625	11.450	3.536	6.718	9.817
1000	0.50	0.10	DML-IVQR-BC	3.900	7.500	10.325	3.538	6.671	9.585
1000	0.75	1.00	Oracle-GMM	0.300	0.300	0.300	0.290	0.290	0.290
1000	0.75	1.00	Full-GMM	0.500	0.500	0.500	0.512	0.512	0.512
1000	0.75	1.00	DML-IVQR-BC	0.300	0.300	0.300	0.279	0.279	0.279
1000	0.75	0.25	Oracle-GMM	2.825	4.200	4.850	2.628	3.945	4.990
1000	0.75	0.25	Full-GMM	2.950	4.450	5.150	2.827	4.267	5.478
1000	0.75	0.25	DML-IVQR-BC	2.325	3.300	3.900	2.402	3.595	4.788
1000	0.75	0.10	Oracle-GMM	3.975	7.750	11.700	3.670	6.999	10.357
1000	0.75	0.10	Full-GMM	3.900	7.750	11.500	3.661	7.023	10.267
1000	0.75	0.10	DML-IVQR-BC	3.900	7.350	10.375	3.554	6.691	9.736
1000	0.90	1.00	Oracle-GMM	0.500	0.500	0.500	0.519	0.519	0.519
1000	0.90	1.00	Full-GMM	1.000	1.000	1.000	1.026	1.031	1.031
1000	0.90	1.00	DML-IVQR-BC	0.400	0.400	0.400	0.454	0.454	0.454
1000	0.90	0.25	Oracle-GMM	3.650	5.850	7.850	3.381	5.625	7.675
1000	0.90	0.25	Full-GMM	3.675	6.150	8.575	3.555	6.123	8.432
1000	0.90	0.25	DML-IVQR-BC	3.450	5.400	7.300	3.114	5.249	7.374
1000	0.90	0.10	Oracle-GMM	4.000	7.900	11.800	3.712	7.241	10.771
1000	0.90	0.10	Full-GMM	3.900	7.700	11.500	3.817	7.448	11.031
1000	0.90	0.10	DML-IVQR-BC	4.000	7.700	11.600	3.644	7.042	10.395

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.2. Grid-domain diagnostic: added accepted measure

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	2.000	1.974	2.000	1.844
1000	0.10	0.25	Full-GMM	2.500	2.470	2.200	2.263
1000	0.10	0.25	DML-IVQR-BC	2.150	2.433	2.000	2.205
1000	0.10	0.10	Oracle-GMM	4.000	3.636	4.000	3.621
1000	0.10	0.10	Full-GMM	3.850	3.610	3.875	3.580
1000	0.10	0.10	DML-IVQR-BC	3.700	3.374	3.450	3.120
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.275	0.933	0.000	0.806
1000	0.25	0.25	Full-GMM	1.025	1.251	0.600	0.978
1000	0.25	0.25	DML-IVQR-BC	1.750	1.500	2.000	1.582
1000	0.25	0.10	Oracle-GMM	3.900	3.207	3.900	3.174
1000	0.25	0.10	Full-GMM	3.800	3.296	3.800	3.192
1000	0.25	0.10	DML-IVQR-BC	3.950	3.275	3.625	3.058
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.000	0.722	0.000	0.547
1000	0.50	0.25	Full-GMM	0.100	0.803	0.000	0.630
1000	0.50	0.25	DML-IVQR-BC	0.250	0.895	0.400	0.921
1000	0.50	0.10	Oracle-GMM	3.900	3.119	4.000	3.047
1000	0.50	0.10	Full-GMM	3.900	3.182	3.900	3.099
1000	0.50	0.10	DML-IVQR-BC	3.850	3.134	2.850	2.914
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	1.325	1.317	0.400	1.045
1000	0.75	0.25	Full-GMM	1.375	1.440	0.900	1.211
1000	0.75	0.25	DML-IVQR-BC	0.525	1.193	1.200	1.192
1000	0.75	0.10	Oracle-GMM	4.000	3.329	4.000	3.358
1000	0.75	0.10	Full-GMM	3.900	3.362	3.800	3.244
1000	0.75	0.10	DML-IVQR-BC	3.575	3.136	3.600	3.045
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.005	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	2.050	2.244	2.000	2.050

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TableC.2. Grid-domain diagnostic: added accepted measure (Continued)

n	τ	κ	**Estimator**	**Med.** $A_0 \to A_1$	**Mean** $A_0 \to A_1$	**Med.** $A_1 \to A_2$	**Mean** $A_1 \to A_2$
1000	0.90	0.25	Full-GMM	2.600	2.568	2.200	2.309
1000	0.90	0.25	DML-IVQR-BC	2.000	2.135	2.000	2.125
1000	0.90	0.10	Oracle-GMM	4.000	3.529	4.000	3.530
1000	0.90	0.10	Full-GMM	3.800	3.631	3.825	3.583
1000	0.90	0.10	DML-IVQR-BC	4.000	3.399	4.000	3.353

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.3. Grid-domain diagnostic: boundary and full-grid acceptance

n	τ	κ	**Estimator**	**Boundary** A_0	**Boundary** A_1	**Boundary** A_2	**All** A_0	**All** A_1	**All** A_2
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.10	0.25	Oracle-GMM	0.880	0.770	0.740	0.150	0.070	0.050
1000	0.10	0.25	Full-GMM	0.920	0.840	0.750	0.090	0.030	0.020
1000	0.10	0.25	DML-IVQR-BC	0.950	0.980	0.980	0.240	0.060	0.050
1000	0.10	0.10	Oracle-GMM	1.000	1.000	1.000	0.580	0.430	0.390
1000	0.10	0.10	Full-GMM	0.990	0.980	0.990	0.430	0.210	0.150
1000	0.10	0.10	DML-IVQR-BC	1.000	0.990	0.990	0.560	0.260	0.200
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.400	0.360	0.270	0.070	0.050	0.030
1000	0.25	0.25	Full-GMM	0.560	0.390	0.390	0.060	0.030	0.010
1000	0.25	0.25	DML-IVQR-BC	0.610	0.740	0.740	0.090	0.050	0.020
1000	0.25	0.10	Oracle-GMM	0.940	0.930	0.940	0.470	0.380	0.320
1000	0.25	0.10	Full-GMM	0.960	0.930	0.900	0.320	0.250	0.210
1000	0.25	0.10	DML-IVQR-BC	0.980	0.990	0.990	0.550	0.400	0.290
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.390	0.200	0.150	0.050	0.040	0.020
1000	0.50	0.25	Full-GMM	0.400	0.240	0.260	0.050	0.020	0.020
1000	0.50	0.25	DML-IVQR-BC	0.440	0.340	0.500	0.060	0.020	0.020
1000	0.50	0.10	Oracle-GMM	0.910	0.890	0.830	0.420	0.350	0.320
1000	0.50	0.10	Full-GMM	0.940	0.890	0.850	0.450	0.320	0.280
1000	0.50	0.10	DML-IVQR-BC	0.900	0.930	0.960	0.450	0.360	0.300
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.75	0.25	Oracle-GMM	0.620	0.470	0.370	0.070	0.050	0.040
1000	0.75	0.25	Full-GMM	0.650	0.550	0.530	0.080	0.010	0.000
1000	0.75	0.25	DML-IVQR-BC	0.550	0.450	0.550	0.090	0.060	0.030
1000	0.75	0.10	Oracle-GMM	0.940	0.960	0.930	0.500	0.400	0.380
1000	0.75	0.10	Full-GMM	0.980	0.920	0.920	0.420	0.270	0.220
1000	0.75	0.10	DML-IVQR-BC	0.950	0.970	0.980	0.470	0.360	0.310
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000	0.000
1000	0.90	0.25	Oracle-GMM	0.850	0.760	0.760	0.250	0.090	0.060

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TableC.3. Grid-domain diagnostic: boundary and full-grid acceptance (Continued)

n	τ	κ	**Estimator**	**Boundary** A_0	**Boundary** A_1	**Boundary** A_2	**All** A_0	**All** A_1	**All** A_2
1000	0.90	0.25	Full-GMM	0.960	0.800	0.820	0.110	0.060	0.030
1000	0.90	0.25	DML-IVQR-BC	0.820	0.770	0.790	0.210	0.100	0.060
1000	0.90	0.10	Oracle-GMM	0.980	1.000	1.000	0.620	0.450	0.430
1000	0.90	0.10	Full-GMM	1.000	0.990	0.990	0.380	0.110	0.060
1000	0.90	0.10	DML-IVQR-BC	0.990	0.980	0.990	0.510	0.394	0.343

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.4. Grid-domain diagnostic: profiles and numerical errors

n	τ	κ	**Estimator**	R	**Succ.** A_0	**Succ.** A_1	**Succ.** A_2	**Fail** A_0	**Fail** A_1	**Fail** A_2	**Stage 1**	**Stage 2**
1000	0.10	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.10	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.25	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.50	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.75	0.10	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	1.00	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	Oracle-GMM	100	100	100	100	0	0	0	0	0

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TableC.4. Grid-domain diagnostic: profiles and numerical errors (Continued)

n	τ	κ	**Estimator**	R	**Succ.** A_0	**Succ.** A_1	**Succ.** A_2	**Fail** A_0	**Fail** A_1	**Fail** A_2	**Stage 1**	**Stage 2**
1000	0.90	0.25	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.25	DML-IVQR-BC	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Oracle-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	Full-GMM	100	100	100	100	0	0	0	0	0
1000	0.90	0.10	DML-IVQR-BC	100	100	99	99	0	1	1	1	0

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Table C.5. Grid-domain diagnostic: stabilization within investigated domains

n	τ	κ	**Estimator**	**Investigated-domain status**
1000	0.10	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.10	1.00	Full-GMM	near-stable within investigated domains
1000	0.10	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.10	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.25	Full-GMM	not stabilized within investigated domains
1000	0.10	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.10	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.10	0.10	Full-GMM	not stabilized within investigated domains
1000	0.10	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.25	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.25	1.00	Full-GMM	near-stable within investigated domains
1000	0.25	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.25	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.25	Full-GMM	not stabilized within investigated domains
1000	0.25	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.25	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.25	0.10	Full-GMM	not stabilized within investigated domains
1000	0.25	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.50	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.50	1.00	Full-GMM	near-stable within investigated domains
1000	0.50	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.50	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.25	Full-GMM	not stabilized within investigated domains
1000	0.50	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.50	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.50	0.10	Full-GMM	not stabilized within investigated domains
1000	0.50	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.75	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.75	1.00	Full-GMM	near-stable within investigated domains
1000	0.75	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.75	0.25	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.25	Full-GMM	not stabilized within investigated domains
1000	0.75	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.75	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.75	0.10	Full-GMM	not stabilized within investigated domains
1000	0.75	0.10	DML-IVQR-BC	not stabilized within investigated domains
1000	0.90	1.00	Oracle-GMM	near-stable within investigated domains
1000	0.90	1.00	Full-GMM	near-stable within investigated domains
1000	0.90	1.00	DML-IVQR-BC	near-stable within investigated domains
1000	0.90	0.25	Oracle-GMM	not stabilized within investigated domains

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TableC.5. Grid-domain diagnostic: stabilization within investigated domains (Continued)

n	τ	κ	Estimator	Investigated-domain status
1000	0.90	0.25	Full-GMM	not stabilized within investigated domains
1000	0.90	0.25	DML-IVQR-BC	not stabilized within investigated domains
1000	0.90	0.10	Oracle-GMM	not stabilized within investigated domains
1000	0.90	0.10	Full-GMM	not stabilized within investigated domains
1000	0.90	0.10	DML-IVQR-BC	not stabilized within investigated domains

Note: Domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. Diagnostics concern only these investigated numerical domains and do not establish global boundedness, unboundedness, or disconnected components.

Appendix D Coverage Forensic Audit

Table D.1. Full empirical coverage and size

n	τ	κ	Estimator	R	Coverage	Size	MCSE	MC 95% interval	.95 inside
500	0.10	1.00	Oracle-GMM	500	0.858	0.142	0.016	[0.825, 0.886]	No
500	0.10	1.00	Full-GMM	500	0.442	0.558	0.022	[0.399, 0.486]	No
500	0.10	1.00	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.10	0.50	Oracle-GMM	500	0.904	0.096	0.013	[0.875, 0.927]	No
500	0.10	0.50	Full-GMM	500	0.816	0.184	0.017	[0.780, 0.848]	No
500	0.10	0.50	DML-IVQR-BC	500	0.928	0.072	0.012	[0.902, 0.948]	No
500	0.10	0.25	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.10	0.25	Full-GMM	500	0.928	0.072	0.012	[0.902, 0.948]	No
500	0.10	0.25	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.10	0.10	Oracle-GMM	500	0.954	0.046	0.009	[0.932, 0.969]	Yes
500	0.10	0.10	Full-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.10	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.25	1.00	Oracle-GMM	500	0.886	0.114	0.014	[0.855, 0.911]	No
500	0.25	1.00	Full-GMM	500	0.788	0.212	0.018	[0.750, 0.822]	No
500	0.25	1.00	DML-IVQR-BC	500	0.976	0.024	0.007	[0.959, 0.986]	No
500	0.25	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.25	0.50	Full-GMM	500	0.920	0.080	0.012	[0.893, 0.941]	No
500	0.25	0.50	DML-IVQR-BC	500	0.960	0.040	0.009	[0.939, 0.974]	Yes
500	0.25	0.25	Oracle-GMM	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.25	0.25	Full-GMM	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.25	0.25	DML-IVQR-BC	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
500	0.25	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
500	0.25	0.10	Full-GMM	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.25	0.10	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.50	1.00	Oracle-GMM	500	0.908	0.092	0.013	[0.879, 0.930]	No
500	0.50	1.00	Full-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
500	0.50	1.00	DML-IVQR-BC	500	0.922	0.078	0.012	[0.895, 0.942]	No
500	0.50	0.50	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.50	Full-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.50	0.50	DML-IVQR-BC	500	0.966	0.034	0.008	[0.946, 0.979]	Yes
500	0.50	0.25	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.50	0.25	Full-GMM	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
500	0.50	0.25	DML-IVQR-BC	500	0.966	0.034	0.008	[0.946, 0.979]	Yes
500	0.50	0.10	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.50	0.10	DML-IVQR-BC	500	0.958	0.042	0.009	[0.937, 0.972]	Yes
500	0.75	1.00	Oracle-GMM	500	0.912	0.088	0.013	[0.884, 0.934]	No
500	0.75	1.00	Full-GMM	500	0.832	0.168	0.017	[0.797, 0.862]	No
500	0.75	1.00	DML-IVQR-BC	500	0.770	0.230	0.019	[0.731, 0.805]	No
500	0.75	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
500	0.75	0.50	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.50	DML-IVQR-BC	500	0.926	0.074	0.012	[0.900, 0.946]	No
500	0.75	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.75	0.25	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.25	DML-IVQR-BC	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.75	0.10	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
500	0.75	0.10	Full-GMM	500	0.956	0.044	0.009	[0.934, 0.971]	Yes
500	0.75	0.10	DML-IVQR-BC	500	0.938	0.062	0.011	[0.913, 0.956]	Yes
500	0.90	1.00	Oracle-GMM	500	0.882	0.118	0.014	[0.851, 0.907]	No
500	0.90	1.00	Full-GMM	500	0.556	0.444	0.022	[0.512, 0.599]	No
500	0.90	1.00	DML-IVQR-BC	500	0.658	0.342	0.021	[0.615, 0.698]	No
500	0.90	0.50	Oracle-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.90	0.50	Full-GMM	500	0.840	0.160	0.016	[0.805, 0.870]	No
500	0.90	0.50	DML-IVQR-BC	500	0.898	0.102	0.014	[0.868, 0.922]	No
500	0.90	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
500	0.90	0.25	Full-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
500	0.90	0.25	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
500	0.90	0.10	Oracle-GMM	500	0.950	0.050	0.010	[0.927, 0.966]	Yes
500	0.90	0.10	Full-GMM	500	0.944	0.056	0.010	[0.920, 0.961]	Yes
500	0.90	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.10	1.00	Oracle-GMM	500	0.860	0.140	0.016	[0.827, 0.888]	No
1000	0.10	1.00	Full-GMM	500	0.502	0.498	0.022	[0.458, 0.546]	No

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
1000	0.10	1.00	DML-IVQR-BC	500	0.968	0.032	0.008	[0.949, 0.980]	Yes
1000	0.10	0.50	Oracle-GMM	500	0.926	0.074	0.012	[0.900, 0.946]	No
1000	0.10	0.50	Full-GMM	500	0.804	0.196	0.018	[0.767, 0.836]	No
1000	0.10	0.50	DML-IVQR-BC	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.10	0.25	Oracle-GMM	500	0.930	0.070	0.011	[0.904, 0.949]	No
1000	0.10	0.25	Full-GMM	500	0.932	0.068	0.011	[0.906, 0.951]	Yes
1000	0.10	0.25	DML-IVQR-BC	500	0.950	0.050	0.010	[0.927, 0.966]	Yes
1000	0.10	0.10	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.10	0.10	Full-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.10	0.10	DML-IVQR-BC	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.25	1.00	Oracle-GMM	500	0.874	0.126	0.015	[0.842, 0.900]	No
1000	0.25	1.00	Full-GMM	500	0.790	0.210	0.018	[0.752, 0.823]	No
1000	0.25	1.00	DML-IVQR-BC	500	0.894	0.106	0.014	[0.864, 0.918]	No
1000	0.25	0.50	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
1000	0.25	0.50	Full-GMM	500	0.908	0.092	0.013	[0.879, 0.930]	No
1000	0.25	0.50	DML-IVQR-BC	500	0.952	0.048	0.010	[0.930, 0.968]	Yes
1000	0.25	0.25	Oracle-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.25	0.25	Full-GMM	500	0.956	0.044	0.009	[0.934, 0.971]	Yes
1000	0.25	0.25	DML-IVQR-BC	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
1000	0.25	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.25	0.10	Full-GMM	500	0.962	0.038	0.009	[0.941, 0.976]	Yes
1000	0.25	0.10	DML-IVQR-BC	500	0.954	0.046	0.009	[0.932, 0.969]	Yes
1000	0.50	1.00	Oracle-GMM	500	0.872	0.128	0.015	[0.840, 0.898]	No
1000	0.50	1.00	Full-GMM	500	0.896	0.104	0.014	[0.866, 0.920]	No
1000	0.50	1.00	DML-IVQR-BC	500	0.776	0.224	0.019	[0.737, 0.810]	No
1000	0.50	0.50	Oracle-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.50	0.50	Full-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.50	0.50	DML-IVQR-BC	500	0.918	0.082	0.012	[0.891, 0.939]	No
1000	0.50	0.25	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.50	0.25	Full-GMM	500	0.936	0.064	0.011	[0.911, 0.954]	Yes
1000	0.50	0.25	DML-IVQR-BC	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.50	0.10	Oracle-GMM	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.50	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.50	0.10	DML-IVQR-BC	500	0.942	0.058	0.010	[0.918, 0.959]	Yes
1000	0.75	1.00	Oracle-GMM	500	0.860	0.140	0.016	[0.827, 0.888]	No
1000	0.75	1.00	Full-GMM	500	0.790	0.210	0.018	[0.752, 0.823]	No
1000	0.75	1.00	DML-IVQR-BC	500	0.632	0.368	0.022	[0.589, 0.673]	No
1000	0.75	0.50	Oracle-GMM	500	0.916	0.084	0.012	[0.888, 0.937]	No
1000	0.75	0.50	Full-GMM	500	0.922	0.078	0.012	[0.895, 0.942]	No
1000	0.75	0.50	DML-IVQR-BC	500	0.842	0.158	0.016	[0.807, 0.871]	No

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TableD.1. Full empirical coverage and size (Continued)

n	τ	κ	**Estimator**	R	**Coverage**	**Size**	**MCSE**	**MC 95% interval**	**.95 inside**
1000	0.75	0.25	Oracle-GMM	500	0.934	0.066	0.011	[0.909, 0.953]	Yes
1000	0.75	0.25	Full-GMM	500	0.928	0.072	0.012	[0.902, 0.948]	No
1000	0.75	0.25	DML-IVQR-BC	500	0.916	0.084	0.012	[0.888, 0.937]	No
1000	0.75	0.10	Oracle-GMM	500	0.946	0.054	0.010	[0.923, 0.963]	Yes
1000	0.75	0.10	Full-GMM	500	0.926	0.074	0.012	[0.900, 0.946]	No
1000	0.75	0.10	DML-IVQR-BC	500	0.930	0.070	0.011	[0.904, 0.949]	No
1000	0.90	1.00	Oracle-GMM	500	0.904	0.096	0.013	[0.875, 0.927]	No
1000	0.90	1.00	Full-GMM	500	0.530	0.470	0.022	[0.486, 0.573]	No
1000	0.90	1.00	DML-IVQR-BC	500	0.498	0.502	0.022	[0.454, 0.542]	No
1000	0.90	0.50	Oracle-GMM	500	0.940	0.060	0.011	[0.916, 0.958]	Yes
1000	0.90	0.50	Full-GMM	500	0.834	0.166	0.017	[0.799, 0.864]	No
1000	0.90	0.50	DML-IVQR-BC	500	0.844	0.156	0.016	[0.810, 0.873]	No
1000	0.90	0.25	Oracle-GMM	500	0.952	0.048	0.010	[0.930, 0.968]	Yes
1000	0.90	0.25	Full-GMM	500	0.920	0.080	0.012	[0.893, 0.941]	No
1000	0.90	0.25	DML-IVQR-BC	500	0.922	0.078	0.012	[0.895, 0.942]	No
1000	0.90	0.10	Oracle-GMM	500	0.964	0.036	0.008	[0.944, 0.977]	Yes
1000	0.90	0.10	Full-GMM	500	0.948	0.052	0.010	[0.925, 0.964]	Yes
1000	0.90	0.10	DML-IVQR-BC	500	0.940	0.060	0.011	[0.916, 0.958]	Yes

Note: Coverage is evaluated directly at $\alpha_0(\tau)$ using the unchanged χ^2_2 cutoff. Intervals are Wilson 95% Monte Carlo intervals.

Appendix E Additional $W_N(a)$ Profiles

Table E.1. Full accepted-set informativeness: measure and grid share

n	τ	κ	**Estimator**	**Med. measure**	**Mean measure**	**Med. share**	**Mean share**	**Mean points**
500	0.10	1.00	Oracle-GMM	0.700	0.736	0.171	0.180	7.37
500	0.10	1.00	Full-GMM	1.600	1.615	0.390	0.395	16.21
500	0.10	1.00	DML-IVQR-BC	2.300	2.303	0.561	0.564	23.13
500	0.10	0.50	Oracle-GMM	2.500	2.437	0.610	0.603	24.70
500	0.10	0.50	Full-GMM	3.050	3.005	0.756	0.745	30.55
500	0.10	0.50	DML-IVQR-BC	3.750	3.576	0.927	0.888	36.43
500	0.10	0.25	Oracle-GMM	3.850	3.497	0.951	0.873	35.78
500	0.10	0.25	Full-GMM	3.700	3.620	0.927	0.904	37.06

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
500	0.10	0.25	DML-IVQR-BC	3.900	3.755	0.976	0.936	38.39
500	0.10	0.10	Oracle-GMM	4.000	3.747	1.000	0.936	38.39
500	0.10	0.10	Full-GMM	3.900	3.766	0.976	0.941	38.60
500	0.10	0.10	DML-IVQR-BC	4.000	3.757	1.000	0.938	38.47
500	0.25	1.00	Oracle-GMM	0.400	0.423	0.098	0.103	4.23
500	0.25	1.00	Full-GMM	0.800	0.824	0.195	0.201	8.24
500	0.25	1.00	DML-IVQR-BC	0.700	0.702	0.171	0.171	7.02
500	0.25	0.50	Oracle-GMM	1.200	1.370	0.293	0.335	13.75
500	0.25	0.50	Full-GMM	2.000	2.120	0.488	0.521	21.36
500	0.25	0.50	DML-IVQR-BC	2.100	2.132	0.512	0.525	21.51
500	0.25	0.25	Oracle-GMM	3.250	3.094	0.805	0.770	31.56
500	0.25	0.25	Full-GMM	3.550	3.342	0.878	0.833	34.13
500	0.25	0.25	DML-IVQR-BC	3.450	3.387	0.854	0.842	34.53
500	0.25	0.10	Oracle-GMM	4.000	3.695	1.000	0.923	37.86
500	0.25	0.10	Full-GMM	3.900	3.708	0.976	0.926	37.97
500	0.25	0.10	DML-IVQR-BC	4.000	3.726	1.000	0.930	38.15
500	0.50	1.00	Oracle-GMM	0.400	0.375	0.098	0.092	3.75
500	0.50	1.00	Full-GMM	0.700	0.676	0.171	0.165	6.76
500	0.50	1.00	DML-IVQR-BC	0.500	0.473	0.122	0.115	4.73
500	0.50	0.50	Oracle-GMM	1.000	1.112	0.244	0.272	11.15
500	0.50	0.50	Full-GMM	1.600	1.715	0.390	0.420	17.22
500	0.50	0.50	DML-IVQR-BC	1.200	1.339	0.293	0.327	13.43
500	0.50	0.25	Oracle-GMM	2.925	2.863	0.732	0.711	29.16
500	0.50	0.25	Full-GMM	3.350	3.185	0.829	0.792	32.47
500	0.50	0.25	DML-IVQR-BC	3.050	2.997	0.756	0.745	30.53
500	0.50	0.10	Oracle-GMM	4.000	3.656	1.000	0.913	37.45
500	0.50	0.10	Full-GMM	3.900	3.670	0.976	0.917	37.59
500	0.50	0.10	DML-IVQR-BC	4.000	3.652	1.000	0.912	37.39
500	0.75	1.00	Oracle-GMM	0.500	0.497	0.122	0.121	4.97
500	0.75	1.00	Full-GMM	1.000	0.978	0.244	0.239	9.78

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	**Estimator**	**Med. measure**	**Mean measure**	**Med. share**	**Mean share**	**Mean points**
500	0.75	1.00	DML-IVQR-BC	0.500	0.495	0.122	0.121	4.95
500	0.75	0.50	Oracle-GMM	1.500	1.676	0.366	0.411	16.86
500	0.75	0.50	Full-GMM	2.400	2.397	0.585	0.590	24.20
500	0.75	0.50	DML-IVQR-BC	1.300	1.489	0.317	0.365	14.98
500	0.75	0.25	Oracle-GMM	3.450	3.241	0.854	0.806	33.06
500	0.75	0.25	Full-GMM	3.650	3.473	0.902	0.865	35.46
500	0.75	0.25	DML-IVQR-BC	3.350	3.089	0.829	0.769	31.52
500	0.75	0.10	Oracle-GMM	4.000	3.731	1.000	0.932	38.21
500	0.75	0.10	Full-GMM	3.900	3.760	0.976	0.939	38.52
500	0.75	0.10	DML-IVQR-BC	4.000	3.661	1.000	0.915	37.50
500	0.90	1.00	Oracle-GMM	1.000	1.131	0.244	0.276	11.31
500	0.90	1.00	Full-GMM	2.000	2.013	0.488	0.494	20.26
500	0.90	1.00	DML-IVQR-BC	0.700	0.793	0.171	0.193	7.93
500	0.90	0.50	Oracle-GMM	3.300	2.944	0.817	0.729	29.89
500	0.90	0.50	Full-GMM	3.350	3.293	0.829	0.818	33.54
500	0.90	0.50	DML-IVQR-BC	2.300	2.347	0.561	0.580	23.78
500	0.90	0.25	Oracle-GMM	3.900	3.617	0.976	0.903	37.00
500	0.90	0.25	Full-GMM	3.800	3.715	0.951	0.928	38.04
500	0.90	0.25	DML-IVQR-BC	3.750	3.401	0.927	0.849	34.80
500	0.90	0.10	Oracle-GMM	4.000	3.779	1.000	0.944	38.72
500	0.90	0.10	Full-GMM	3.900	3.816	0.976	0.954	39.11
500	0.90	0.10	DML-IVQR-BC	4.000	3.704	1.000	0.926	37.95
1000	0.10	1.00	Oracle-GMM	0.400	0.379	0.098	0.092	3.79
1000	0.10	1.00	Full-GMM	0.800	0.766	0.195	0.187	7.66
1000	0.10	1.00	DML-IVQR-BC	0.600	0.668	0.146	0.163	6.68
1000	0.10	0.50	Oracle-GMM	1.000	1.291	0.244	0.316	12.97
1000	0.10	0.50	Full-GMM	1.900	1.941	0.463	0.477	19.56
1000	0.10	0.50	DML-IVQR-BC	2.225	2.283	0.549	0.562	23.04
1000	0.10	0.25	Oracle-GMM	3.450	3.077	0.854	0.765	31.36
1000	0.10	0.25	Full-GMM	3.500	3.318	0.878	0.826	33.87

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
1000	0.10	0.25	DML-IVQR-BC	3.750	3.577	0.927	0.890	36.49
1000	0.10	0.10	Oracle-GMM	4.000	3.740	1.000	0.934	38.30
1000	0.10	0.10	Full-GMM	3.900	3.781	0.976	0.945	38.73
1000	0.10	0.10	DML-IVQR-BC	4.000	3.826	1.000	0.955	39.15
1000	0.25	1.00	Oracle-GMM	0.200	0.249	0.049	0.061	2.49
1000	0.25	1.00	Full-GMM	0.400	0.433	0.098	0.106	4.33
1000	0.25	1.00	DML-IVQR-BC	0.300	0.341	0.073	0.083	3.41
1000	0.25	0.50	Oracle-GMM	0.600	0.659	0.146	0.161	6.59
1000	0.25	0.50	Full-GMM	1.000	1.020	0.244	0.249	10.20
1000	0.25	0.50	DML-IVQR-BC	0.800	0.915	0.195	0.223	9.16
1000	0.25	0.25	Oracle-GMM	2.200	2.271	0.537	0.561	23.02
1000	0.25	0.25	Full-GMM	2.650	2.611	0.659	0.647	26.51
1000	0.25	0.25	DML-IVQR-BC	2.850	2.669	0.707	0.661	27.09
1000	0.25	0.10	Oracle-GMM	3.900	3.544	0.976	0.885	36.27
1000	0.25	0.10	Full-GMM	3.900	3.627	0.976	0.906	37.14
1000	0.25	0.10	DML-IVQR-BC	3.900	3.637	0.976	0.908	37.22
1000	0.50	1.00	Oracle-GMM	0.200	0.228	0.049	0.056	2.28
1000	0.50	1.00	Full-GMM	0.400	0.369	0.098	0.090	3.69
1000	0.50	1.00	DML-IVQR-BC	0.300	0.261	0.073	0.064	2.61
1000	0.50	0.50	Oracle-GMM	0.500	0.562	0.122	0.137	5.62
1000	0.50	0.50	Full-GMM	0.800	0.833	0.195	0.203	8.33
1000	0.50	0.50	DML-IVQR-BC	0.600	0.641	0.146	0.156	6.41
1000	0.50	0.25	Oracle-GMM	1.900	1.946	0.463	0.480	19.67
1000	0.50	0.25	Full-GMM	2.200	2.277	0.537	0.562	23.03
1000	0.50	0.25	DML-IVQR-BC	2.025	2.068	0.500	0.510	20.93
1000	0.50	0.10	Oracle-GMM	3.900	3.501	0.976	0.874	35.83
1000	0.50	0.10	Full-GMM	3.800	3.520	0.951	0.879	36.02
1000	0.50	0.10	DML-IVQR-BC	3.900	3.514	0.976	0.877	35.96
1000	0.75	1.00	Oracle-GMM	0.300	0.280	0.073	0.068	2.80
1000	0.75	1.00	Full-GMM	0.500	0.518	0.122	0.126	5.18

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TableE.1. Full accepted-set informativeness: measure and grid share (Continued)

n	τ	κ	Estimator	Med. measure	Mean measure	Med. share	Mean share	Mean points
1000	0.75	1.00	DML-IVQR-BC	0.300	0.283	0.073	0.069	2.83
1000	0.75	0.50	Oracle-GMM	0.700	0.830	0.171	0.203	8.31
1000	0.75	0.50	Full-GMM	1.100	1.217	0.268	0.297	12.18
1000	0.75	0.50	DML-IVQR-BC	0.700	0.737	0.171	0.180	7.37
1000	0.75	0.25	Oracle-GMM	2.650	2.488	0.659	0.615	25.22
1000	0.75	0.25	Full-GMM	2.900	2.791	0.707	0.691	28.32
1000	0.75	0.25	DML-IVQR-BC	2.250	2.266	0.561	0.560	22.98
1000	0.75	0.10	Oracle-GMM	3.900	3.609	0.976	0.901	36.92
1000	0.75	0.10	Full-GMM	3.900	3.647	0.976	0.911	37.35
1000	0.75	0.10	DML-IVQR-BC	3.850	3.519	0.951	0.878	36.01
1000	0.90	1.00	Oracle-GMM	0.500	0.515	0.122	0.126	5.15
1000	0.90	1.00	Full-GMM	1.000	1.035	0.244	0.252	10.35
1000	0.90	1.00	DML-IVQR-BC	0.400	0.446	0.098	0.109	4.46
1000	0.90	0.50	Oracle-GMM	1.575	1.864	0.390	0.458	18.78
1000	0.90	0.50	Full-GMM	2.450	2.417	0.610	0.595	24.39
1000	0.90	0.50	DML-IVQR-BC	1.100	1.414	0.268	0.346	14.21
1000	0.90	0.25	Oracle-GMM	3.650	3.279	0.902	0.815	33.42
1000	0.90	0.25	Full-GMM	3.600	3.462	0.902	0.862	35.35
1000	0.90	0.25	DML-IVQR-BC	3.350	2.986	0.829	0.742	30.44
1000	0.90	0.10	Oracle-GMM	4.000	3.690	1.000	0.922	37.80
1000	0.90	0.10	Full-GMM	3.900	3.767	0.976	0.941	38.59
1000	0.90	0.10	DML-IVQR-BC	4.000	3.604	1.000	0.900	36.92

Note: Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$. The measure is finite-domain and is not an unrestricted confidence-region length.

Table E.2. Full accepted-set informativeness: boundary diagnostics, n = 500

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
500	0.10	1.00	Oracle-GMM	0.004	0.000	0.004	0.000	0.000
500	0.10	1.00	Full-GMM	0.058	0.056	0.106	0.008	0.000
500	0.10	1.00	DML-IVQR-BC	0.036	0.180	0.212	0.004	0.000
500	0.10	0.50	Oracle-GMM	0.288	0.376	0.552	0.112	0.038
500	0.10	0.50	Full-GMM	0.482	0.514	0.740	0.256	0.006
500	0.10	0.50	DML-IVQR-BC	0.402	0.926	0.954	0.374	0.216
500	0.10	0.25	Oracle-GMM	0.788	0.824	0.962	0.650	0.370
500	0.10	0.25	Full-GMM	0.838	0.864	0.972	0.730	0.146
500	0.10	0.25	DML-IVQR-BC	0.702	0.980	0.996	0.686	0.440
500	0.10	0.10	Oracle-GMM	0.928	0.926	0.994	0.860	0.564
500	0.10	0.10	Full-GMM	0.936	0.940	0.990	0.886	0.292
500	0.10	0.10	DML-IVQR-BC	0.840	0.952	0.992	0.800	0.514
500	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.25	0.50	Oracle-GMM	0.048	0.060	0.102	0.006	0.000
500	0.25	0.50	Full-GMM	0.168	0.158	0.298	0.028	0.000
500	0.25	0.50	DML-IVQR-BC	0.020	0.354	0.364	0.010	0.004
500	0.25	0.25	Oracle-GMM	0.592	0.652	0.836	0.408	0.190
500	0.25	0.25	Full-GMM	0.684	0.734	0.906	0.512	0.172
500	0.25	0.25	DML-IVQR-BC	0.428	0.906	0.946	0.388	0.204
500	0.25	0.10	Oracle-GMM	0.880	0.924	0.996	0.808	0.514
500	0.25	0.10	Full-GMM	0.876	0.916	0.988	0.804	0.364
500	0.25	0.10	DML-IVQR-BC	0.806	0.966	0.990	0.782	0.502
500	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.50	0.50	Oracle-GMM	0.042	0.018	0.056	0.004	0.000
500	0.50	0.50	Full-GMM	0.096	0.044	0.132	0.008	0.000
500	0.50	0.50	DML-IVQR-BC	0.010	0.072	0.078	0.004	0.000
500	0.50	0.25	Oracle-GMM	0.560	0.490	0.706	0.344	0.168
500	0.50	0.25	Full-GMM	0.644	0.602	0.798	0.448	0.150
500	0.50	0.25	DML-IVQR-BC	0.380	0.742	0.810	0.312	0.150
500	0.50	0.10	Oracle-GMM	0.898	0.868	0.968	0.798	0.540
500	0.50	0.10	Full-GMM	0.884	0.882	0.974	0.792	0.404
500	0.50	0.10	DML-IVQR-BC	0.818	0.936	0.984	0.770	0.540
500	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
500	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
500	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
500	0.75	0.50	Oracle-GMM	0.172	0.030	0.196	0.006	0.000

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TableE.2. Full accepted-set informativeness: boundary diagnostics, $n = 500$ (Continued)

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
500	0.75	0.50	Full-GMM	0.300	0.164	0.394	0.070	0.010
500	0.75	0.50	DML-IVQR-BC	0.076	0.098	0.164	0.010	0.000
500	0.75	0.25	Oracle-GMM	0.750	0.552	0.872	0.430	0.224
500	0.75	0.25	Full-GMM	0.766	0.702	0.916	0.552	0.176
500	0.75	0.25	DML-IVQR-BC	0.552	0.714	0.854	0.412	0.226
500	0.75	0.10	Oracle-GMM	0.906	0.896	0.988	0.814	0.564
500	0.75	0.10	Full-GMM	0.928	0.912	0.992	0.848	0.426
500	0.75	0.10	DML-IVQR-BC	0.842	0.946	0.994	0.794	0.524
500	0.90	1.00	Oracle-GMM	0.012	0.004	0.016	0.000	0.000
500	0.90	1.00	Full-GMM	0.160	0.102	0.246	0.016	0.000
500	0.90	1.00	DML-IVQR-BC	0.002	0.000	0.002	0.000	0.000
500	0.90	0.50	Oracle-GMM	0.558	0.346	0.698	0.206	0.098
500	0.90	0.50	Full-GMM	0.640	0.584	0.834	0.390	0.022
500	0.90	0.50	DML-IVQR-BC	0.294	0.338	0.538	0.094	0.028
500	0.90	0.25	Oracle-GMM	0.872	0.790	0.978	0.684	0.446
500	0.90	0.25	Full-GMM	0.898	0.878	0.996	0.780	0.190
500	0.90	0.25	DML-IVQR-BC	0.738	0.838	0.978	0.598	0.326
500	0.90	0.10	Oracle-GMM	0.938	0.914	0.996	0.856	0.616
500	0.90	0.10	Full-GMM	0.936	0.960	0.996	0.900	0.292
500	0.90	0.10	DML-IVQR-BC	0.884	0.946	0.998	0.832	0.558

Note: Boundary rates refer to the endpoints of $A_0 = [-1, 3]$; 'All' is the rate at which all 41 primary-grid points are accepted.

Table E.3. Full accepted-set informativeness: boundary diagnostics, $n = 1000$

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
1000	0.10	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.10	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.10	0.50	Oracle-GMM	0.050	0.072	0.122	0.000	0.000
1000	0.10	0.50	Full-GMM	0.176	0.132	0.292	0.016	0.002
1000	0.10	0.50	DML-IVQR-BC	0.088	0.332	0.398	0.022	0.006
1000	0.10	0.25	Oracle-GMM	0.556	0.626	0.836	0.346	0.166
1000	0.10	0.25	Full-GMM	0.670	0.714	0.908	0.476	0.092
1000	0.10	0.25	DML-IVQR-BC	0.536	0.904	0.956	0.484	0.260
1000	0.10	0.10	Oracle-GMM	0.862	0.936	0.986	0.812	0.548
1000	0.10	0.10	Full-GMM	0.910	0.934	0.998	0.846	0.358
1000	0.10	0.10	DML-IVQR-BC	0.814	0.968	0.992	0.790	0.528
1000	0.25	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.25	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.25	0.50	Oracle-GMM	0.000	0.004	0.004	0.000	0.000
1000	0.25	0.50	Full-GMM	0.002	0.008	0.010	0.000	0.000
1000	0.25	0.50	DML-IVQR-BC	0.000	0.014	0.014	0.000	0.000
1000	0.25	0.25	Oracle-GMM	0.268	0.340	0.500	0.108	0.052
1000	0.25	0.25	Full-GMM	0.382	0.408	0.606	0.184	0.046
1000	0.25	0.25	DML-IVQR-BC	0.206	0.594	0.652	0.148	0.062
1000	0.25	0.10	Oracle-GMM	0.818	0.840	0.964	0.694	0.454
1000	0.25	0.10	Full-GMM	0.868	0.860	0.974	0.754	0.378
1000	0.25	0.10	DML-IVQR-BC	0.756	0.938	0.978	0.716	0.488
1000	0.50	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.50	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.50	Full-GMM	0.002	0.000	0.002	0.000	0.000
1000	0.50	0.50	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.50	0.25	Oracle-GMM	0.238	0.182	0.342	0.078	0.036
1000	0.50	0.25	Full-GMM	0.286	0.234	0.420	0.100	0.030
1000	0.50	0.25	DML-IVQR-BC	0.146	0.352	0.408	0.090	0.040
1000	0.50	0.10	Oracle-GMM	0.828	0.812	0.916	0.724	0.418
1000	0.50	0.10	Full-GMM	0.824	0.832	0.934	0.722	0.366
1000	0.50	0.10	DML-IVQR-BC	0.740	0.882	0.926	0.696	0.428
1000	0.75	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	Full-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.75	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.75	0.50	Oracle-GMM	0.008	0.000	0.008	0.000	0.000

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TableE.3. Full accepted-set informativeness: boundary diagnostics, $n = 1000$ (Continued)

n	τ	κ	**Estimator**	**Left**	**Right**	**Either**	**Both**	**All**
1000	0.75	0.50	Full-GMM	0.010	0.000	0.010	0.000	0.000
1000	0.75	0.50	DML-IVQR-BC	0.004	0.000	0.004	0.000	0.000
1000	0.75	0.25	Oracle-GMM	0.472	0.222	0.568	0.126	0.058
1000	0.75	0.25	Full-GMM	0.516	0.308	0.628	0.196	0.046
1000	0.75	0.25	DML-IVQR-BC	0.314	0.320	0.498	0.136	0.054
1000	0.75	0.10	Oracle-GMM	0.856	0.818	0.956	0.718	0.458
1000	0.75	0.10	Full-GMM	0.904	0.862	0.974	0.792	0.380
1000	0.75	0.10	DML-IVQR-BC	0.770	0.872	0.960	0.682	0.412
1000	0.90	1.00	Oracle-GMM	0.000	0.000	0.000	0.000	0.000
1000	0.90	1.00	Full-GMM	0.002	0.000	0.002	0.000	0.000
1000	0.90	1.00	DML-IVQR-BC	0.000	0.000	0.000	0.000	0.000
1000	0.90	0.50	Oracle-GMM	0.236	0.044	0.270	0.010	0.004
1000	0.90	0.50	Full-GMM	0.274	0.148	0.374	0.048	0.000
1000	0.90	0.50	DML-IVQR-BC	0.106	0.030	0.134	0.002	0.000
1000	0.90	0.25	Oracle-GMM	0.728	0.534	0.864	0.398	0.232
1000	0.90	0.25	Full-GMM	0.796	0.666	0.940	0.522	0.142
1000	0.90	0.25	DML-IVQR-BC	0.574	0.576	0.808	0.342	0.174
1000	0.90	0.10	Oracle-GMM	0.906	0.886	0.980	0.812	0.580
1000	0.90	0.10	Full-GMM	0.924	0.906	0.994	0.836	0.360
1000	0.90	0.10	DML-IVQR-BC	0.838	0.914	0.986	0.766	0.506

Note: Boundary rates refer to the endpoints of $A_0 = [-1, 3]$; ‘All’ is the rate at which all 41 primary-grid points are accepted.

Appendix F Implementation and Reproducibility

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

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